



Cite this: *Biomater. Sci.*, 2018, **6**, 2812

Received 18th April 2018,
Accepted 18th September 2018

DOI: 10.1039/c8bm00422f

rs.c.li/biomaterials-science

3D scaffolds for brain tissue regeneration: architectural challenges

Gillian Dumsile Mahumane,^{id} Pradeep Kumar,^{id} Lisa Claire du Toit, Yahya Essop Choonara and Viness Pillay^{id} *

Biomaterials are being utilized to engender biomimetic, pro-regenerative constructs in the form of 3D scaffolds to augment functional neural tissue (brain tissue) repair and regeneration. Tissue engineered three-dimensional (3D) scaffolds have shown various degrees of experimental success, indicating great potential for development as treatment options. However, there is yet to be a 3D scaffold that exhibits consummate results of an effective clinical standard. Critical assessment of the performance of current 3D scaffolds could provide insightful feedback for tailoring future 3D scaffolds towards more promising results. This review provides a critical analysis of current 3D scaffolds for neural tissue engineering. Architectural properties, such as porosity, swelling, and architectural influences, such as design approach and polymeric material choice, were scrutinized for suitability for the desired tissue target properties. Success and shortcomings of various 3D scaffolds were evaluated through the analysis of tissue integration of the 3D scaffold *in vivo*. Investigations focused on in this review included those: (1) reporting at an *in vivo* experimental level in animal models, (2) involving polymer-based (natural/synthetic) scaffolds described as possessing a '3D' architecture, (3) targeting brain tissue regeneration (4) published from 2011 onward.

Wits Advanced Drug Delivery Platform Research Unit, Department of Pharmacy and Pharmacology, School of Therapeutic Science, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, 7 York Road, Parktown, 2193, South Africa.
E-mail: viness.pillay@wits.ac.za; Fax: +27-11-642-4355, +27-86-553-4733;
Tel: +27-11-717-2274

1. Introduction

1.1. Regeneration potential of the central nervous system

An understanding of the regenerative potential of neural tissue is crucial to scrutinizing the current three-dimensional (3D) scaffold interventions for neural tissue regeneration. From a



Gillian Dumsile Mahumane

Gillian Dumsile Mahumane is currently a postgraduate student (PhD candidate) at the Wits Advanced Drug Delivery Platform (WADDP), University of the Witwatersrand, Johannesburg, South Africa. Her current project research areas include tissue engineering, 3D biomaterial scaffolds, nanotechnology, neural tissue research and pharmaceuticals. She has a Bachelor of Pharmacy undergraduate degree and a Master of Pharmacy (with distinction) postgraduate degree (both from the University of the Witwatersrand).



Pradeep Kumar

Pradeep Kumar (BPharm, MPharm, PhD) is a Senior Lecturer of Pharmaceutics at Wits and a Senior Researcher at the Wits Advanced Drug Delivery Platform. Pradeep's doctoral thesis was focused on engineering neural devices for spinal cord injury interventions and was awarded the Most Prestigious PhD Degree Award for his research. In addition, he has established a computational analysis and programming algorithm for the design of drug delivery devices and polymeric architectures. Dr Kumar has contributed to over 185 publications, 4 editorials, 3 edited books, 26 book chapters, and 6 granted/12 filed patents (Google Scholar H-index: 25).

macro/tissue perspective, the consequence of neurotrauma is a cavitory defect/some form of tissue discontinuity at the lesion site and a disruption in the unidirectional architecture of the nerve tracts.^{1–4} This discontinuity results in a destruction of the neural network that transmits signals responsible for so many body functions in the central nervous system (CNS). Neural network repair may be crucial to CNS function. This can only happen if there is functional tissue continuity at the cavitory site. Naturally, 'bridging' the lesion is the logical step. The challenge in this step from a tissue engineering perspective has been ensuring that the 'bridge' (scaffold) mimics the native tissue as closely as possible, biologically and architecturally.

From a micro/cellular perspective, nerve injury results in a loss of neurons and/or supporting cells.⁴ Generation of new neurons is limited to specified areas, such as the sub-ventricular zone and the dentate gyrus.^{5–7} Mature neurons are post-mitotic, meaning they do not divide to give rise to new cells.⁸ Instead, they may be generated from neural stem cells or *via in situ* reprogramming of endogenous astrocytes to neurons.^{2,6} Neural stem cells are able to both self-renew and differentiate into neurons, astrocytes and oligodendrocytes.^{4,5} Su and co-workers² showed that astrocytes can be converted into doublecortin (DCX)-positive neuroblasts which could mature into synapse-forming neurons *in vivo* in the spinal cord by a single transcription factor, sex-determining region Y-box 11 (SOX11). These astrocytes can be reprogrammed in the brain as well. Provided with a favourable cellular milieu, neurons can regenerate their axons post injury.⁶ In general, the peripheral nervous system (PNS) shows better recovery in comparison to the CNS post injury. The difference in results has in part been linked to cellular composition, and the presence and absence of Schwann cells in the PNS and CNS respectively. Schwann cells guide, myelinate nerve fibres, secrete neurotrophins and produce extracellular matrix (ECM) molecules, functions

which facilitate neural regeneration.^{4–6} Whereas in the CNS, astrocytes contribute to scar formation and oligodendrocytes express inhibitory proteins in the area, functions which hinder axonal outgrowth and neural regeneration effort.^{4,8,9} This difference in neuroglia between the PNS and the CNS is one of main differences in the regeneration potential between the two. In both instances, regeneration can occur; the requirements from a tissue perspective are slightly varied in focus keeping in mind the cell types of the PNS and CNS, respectively. With the PNS, a guidance structure is the main requirement for regeneration. Whereas for the CNS, in addition to guidance, the structure needs to be able to alter the inhibitory environment by exhibiting functions similar to the Schwann cells.

1.2. Therapeutic interventions for neurotrauma

Stem cell transplantation alone or in combination with neurotrophins have been introduced with the goal that they may replace lost cells in the injured area and stimulate regeneration.^{5,10–12} These cells can be genetically modified to over-express neurotrophic factors such as brain-derived neurotrophic factor (BDNF), which aids in the growth of axons.¹³ The shortfall with the cell/neurotrophin strategy is the lack of a physical 3D construct at the lesion site to facilitate directed axonal growth of the stimulated axons. Axons tend to grow in an arbitrary orientation and thus very few reach the distal portion of the lesion for reconnection of the neural circuit.¹³ Years of cell transplants have shown that cells show poor survival when presented as dissociated suspensions in comparison to implantation as donor tissue or when presented within scaffolds, in part due to exposure to and lack of protection from the inhibitory environment at the lesion site.^{4,9,13–18} Losses as high as 90–99% of transplanted cells have been reported.¹⁹



Lisa Claire du Toit

Lisa Claire du Toit, B.Pharm, M.Pharm, PhD (Wits) is currently an Associate Professor of Pharmaceutical Chemistry and Senior Researcher of the Wits Advanced Drug Delivery Platform Research Unit (WADDP) in the Department of Pharmacy and Pharmacology, University of the Witwatersrand, Johannesburg, South Africa. Prof. du Toit has over 170 publications which have been published in high impact factor, reputable ISI

recognised journals, with most of these journals falling within the first quartile in Pharmaceutical Sciences. She has an h-index of 26 (Google Scholar). Prof. du Toit is the first author of 4 book chapters and has co-authored a number of editorials, books or chapters in books.



Yahya Essop Choonara

Yahya E. Choonara (PhD) is Head of Pharmacy and Pharmacology at Wits University. He is a Personal Professor of Pharmaceutics and Senior Scientist of the Wits Advanced Drug Delivery Platform (WADDP) Research Unit. He has made significant contributions to optimizing drug delivery technologies using PharMATHaceutics to deduce and optimize the mechanisms of drug release from polymer-engineered systems. His

researcher standing is reflected in his publication record of >230 articles in ISI-accredited journals having >4100 citations with an H-index of 32. He has filed 43 PCT patents and received numerous international awards as a prolific inventor.

The alternative to the transplantation of molecules and stem cells are 'surgical' and 'bridging' treatment options. End-to-end suturing is the gold standard for surgical treatment.^{19,20} This treatment option is limited to ideal settings, such as: clean-cut nerve transaction, minimal tissue loss, denervation of less than 6 months, minute gaps (<1 cm (ref. 21 and 22)) and an injury zone with good blood supply surrounded by soft tissue.^{19,20} Unfortunately, injuries vary and 'ideal' lesions are not representative of the clinical reality.

For non-ideal lesions (e.g. with bigger nerve gaps ranging from 1 to 5 cm (ref. 22)), a 'bridging' treatment option is often required.^{4,23} The "gold-standard" treatment approach for nerve injuries (e.g. PNS) is an autologous nerve transplant therapy when the gap considered is 'big'.^{4,6,19,20,24–28} Autologous nerve transplants are, however, limited by their zero-sum nature as therapeutic options (requiring sacrifice of healthy donor tissue) that does not necessarily guarantee a gain. For example, this therapy option results in donor-site morbidity, mismatched size of graft and recipient nerve further complicated by processes such as intraneural fibrosis, scar formation, and vascular complications.^{4,6,24–26,29,30} These all lead to narrowed chances of therapeutic success and neural regeneration. Alternate bridging material, such as allografts, present limitations such as a high risk of immune-rejection.²⁰

To avoid the zero-sum nature of autologous grafts and host rejection of allografts, various artificial scaffolds have been developed as alternative bridging material. Scaffolds are systems used to re-establish tissue continuity at the cavitary site.¹ They promise a non-zero-sum therapeutic alternative (requiring no sacrifice of healthy donor tissue) by serving as a temporary stand-in structure to stimulate and support the

reconstruction and repair of nerve tissue. The tissue continuity offered at the lesion site varies with each scaffold system (two-dimensional (2D) *versus* 3D).

A few neural scaffolds have attained Food and Drug Administration (FDA) approval (e.g. Neuragen®, Neuroflex™, NeuroMatrix™ (collagen nerve guidance conduits); NeuroTube® polyglycolic acid (PGA) nerve guidance conduit; Neurolac™ poly(D,L-lactide-epsilon-caprolactone) (PCL) nerve guidance conduit).³¹ Even these FDA-approved scaffolds show clinical limitations in that their performance is generally limited to 3 cm gaps and are generally inferior to the performance of autograft treatment.^{4,19,20,25,26,31,32} For example, Neurolac is extremely rigid resulting in breaking of needles and nerve tearing during implantation, suturing and regeneration, respectively.³³ The majority of these FDA-approved/commercial scaffolds are mainly for application in the PNS.⁴ Far less success is noted with CNS research as scaffolds for CNS repair are largely in the experimental/research phase. Furthermore, current treatment approaches for conditions of the brain such as traumatic brain injury (TBI), are largely focused on preventing further damage (pharmacological agents and rehabilitation) rather than facilitating regeneration.¹⁰

1.3. Trends in 3D scaffolding for neural tissue engineering/application

There has been a noteworthy growth in the design approach to 3D scaffolding. Neural tissue engineering efforts have gravitated towards a novel type of neural scaffold that addresses the shortcomings of previous scaffolds while probing at fundamental concepts behind neuro-mimicry. Kumar and co-workers³⁴ suggested 'a multifunctional combinatorial device assimilating the biomaterial, cellular, molecular, structural and functional aspects altogether is the best way forward for effective neuroregeneration post traumatic spinal cord injury'. This observation is applicable across the board for neural tissue engineering. Fig. 1 illustrates the new type of scaffold design approach observed in recent (from 2011 onward) literature. It is multimodal, encompassing this 'multifunctional combinatorial' nature.

Advances in technology such as computer aided design have led to promising efforts towards specialized architecture. Accompanying these potential design approaches are the improvements in fabrication methods, such as application of 3D bioprinting. Furthermore, there have been great efforts in terms of optimizing material properties. These advances have resulted in a variety of 3D biomaterial scaffolds, with diverse architectural features for application in neural tissue regeneration (Table 1).

However, there are still shortfalls regarding the performance of these 3D scaffolds *in vivo*. The majority of the work is largely experimental with very little investigations performed at an *in vivo* level. Furthermore, hardly any scaffolds reach clinical testing stages, especially with regards to CNS regeneration in traumatic brain injury.³⁶ Most of the scaffolds are reported as possessing 'potential' for neural tissue regeneration applications.^{36–38} These observations surrounding neural



Viness Pillay

Viness Pillay, a Fulbright Scholar and Fellow of the African Academy of Sciences (AAS), is a South African National Research Foundation (NRF) Research Chair in Pharmaceutical Biomaterials and Polymer-Engineered Drug Delivery Technologies hosted by the University of the Witwatersrand, funded by the Department of Science and Technology (DST) and administered by the NRF at the Tier 1

Level. Prof. Pillay is a Personal Professor of Pharmaceutics, Head of Pharmaceutics and Director of the Wits Advanced Drug Delivery Platform (WADDP) Research Unit and Contract Research at the University of the Witwatersrand, Department of Pharmacy and Pharmacology, South Africa. His research Unit encompasses the research, development and commercialization of innovative drug delivery systems currently under development by his research team.

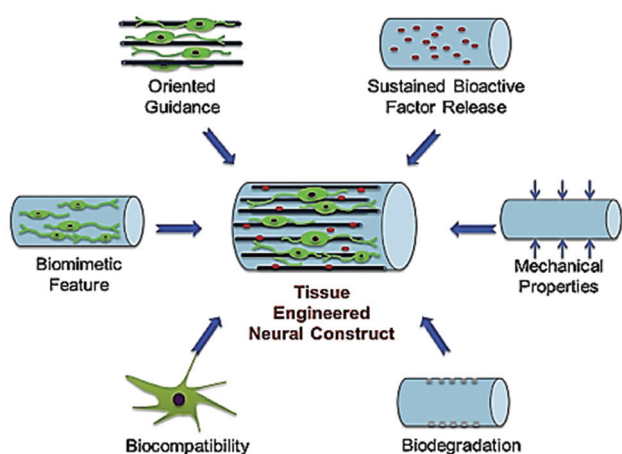


Fig. 1 Parameters governing recent multifunctional 3D neural scaffold designs. This figure is reused from *Nanomedicine: Nanotechnology, Biology, and Medicine*, 11, Zhu and co-workers.³⁵ Highly aligned nanocomposite scaffolds by electrospinning and electrospraying for neural tissue regeneration, pp. 693–704, 2015, with permission from Elsevier.

tissue regeneration are an indicator of the need for a critical review of the success and shortcomings of 3D scaffolds to consolidate their true potential for the desired function (Fig. 2).

2. Neural tissue-based critical architectural design challenges for neuromimicry in 3D scaffolds

Biomimicry requires modelling the 3D scaffold on the biological target tissue processes, properties and features. The current challenge of biomimicry is amplified by the incomplete understanding of the following properties of brain tissue as a biological system:

- **Heterogeneity:** Neural tissue is comprised of varied anatomical regions due to the varied density and alignment of the tissues components (e.g. different cell types, biomolecules, blood vessels, fibrous proteins) in various regions. This innate tissue heterogeneity points to the functional importance of niche areas/pockets which poses a translational challenge in biomimetic 3D architectural design when there is an incomplete understanding of the tissue as a system.

- **Complex 3D tissue:** Ultimately, 3D culturing is a process crucial to reconstructing the *in vivo* structure-function relationship of the neural tissue.⁴⁹ This would ensure that the newly formed tissue is 3D to enhance integration with the surrounding 3D tissue at the lesion zone. However, replicating the structural and functional framework has thus far been a challenge due to the high complexity of the 3D neural tissue network.

- **Unique neuronal micro-environment:** *In vivo*, the immediate microenvironment of neural cells is composed of the interstitial system, which accounts for approximately 20% of the total brain volume.^{49–51} The interstitial system (ISS) or neural micro-environment comprises of the extracellular matrix and the interstitial fluid (ISF). Briefly, the interstitial fluid serves as a

medium that bathes the cells in a rich environment (of for example, contains hormones, neurotransmitters) and facilitates waste removal, nutrient supply, organic molecules such as proteins; while the extracellular matrix directly attaches to and anchors the cells as depicted in Fig. 4a and b.^{51,52} Fig. 4b specifically highlights that the ECM enables intercellular adhesion and functions as a scaffold.⁵² The native ECM presents a macro to nanoscale topography, therefore, 3D scaffolds need to be processed to exhibit similar topographical cues (e.g. fibres and pores). Neural tissue has a dense vasculature in order to facilitate mass transport (e.g. of cells, nutrients, waste).⁵³ Interconnected pores and channels designed in the architecture can accommodate for the absence of vasculature at the lesion site by facilitating mass transport within the 3D scaffolds.

2.1. Macro-architectural profile of 3D scaffolds: biomimicry of tissue properties, and translational potential

Biomimicry is an important conceptualization point for the architectural design of a scaffold for brain tissue repair. The macroscopic architecture of the brain consists of a solid phase (*i.e.* brain tissue) and a fluid phase.^{53,54} Terms such as ‘porous fluid-saturated solid’ and ‘fluid-bathed multiphase material’ have been employed to describe the properties of brain tissue.^{55,56} Table 1 and Fig. 3 represented the properties of the various 3D scaffold designs developed during the period from 2011 onward for brain implantation. The type of scaffold systems noted are herein categorized into two major groups based on water content; solid systems (e.g. sponges, nanofibers) and liquid systems (hydrogels). Solid systems possess a very low water content (‘dry’) in comparison to the ‘fluid-bathed’ nature of brain tissue. Moreover the ‘dry’ scaffold at a lesion cavity serves as an inferior media for cell support in terms of the lack of an environment that enables diffusion of nutrients, oxygen and bioactive factors for cell survival and growth. Hybrids of these scaffold systems are also noted (e.g. electrospun fibres dispersed within hydrogel). The “porous fluid-saturated solid” property used to describe brain tissue may serve as the rationale behind the combination and prevalence of these two types of systems in literature. These scaffold systems are largely integrative and supportive^{11,39} due to the unique integration of common architectural features such as interconnected pores (with an average pore size ranging from 20–160 μm),^{37,39,44} channels^{38,57} and electrospun fibres⁴⁵ (Table 1).

Literature (Table 1) reveals fabricated 3D architectures with an assortment of uniform and non-uniform architectures, hydrogels and nanofibers, various sizes of channels and pores, and various architectural compositions (polymeric biomaterial, neurotropic factors, cell adhesive proteins and peptides and cell seeding). This heterogeneity presents an assortment of data for analysis, thus rendering comparison of said data highly difficult due to innate differences in 3D architectures and study designs. Therefore, this review is limited by the lack of comparability of most investigations in this field. In light of this, the choice to review literature at an *in vivo* level enables the institution of performance (e.g. glial scar formation, cell

Table 1 Recent 3D scaffolds and their outcomes following *in vivo* assessment investigated for the repair of brain tissue (2011-onwards)

Reference and publication date of work	Scaffold system/type	Scaffold fabrication methods and resultant architectural features	Cell-seeding/encapsulation	<i>In vivo</i> animal model and number of animals tested and duration of implant analysis	Effect of scaffold at lesion site	Functional behavioural assessment and improvement
Jurga and co-workers, 2011 ³⁹	Gelatin crosslinked with laminin based cryogel	Cryogelation methods yielded a macroporous sponge-like cryogel with pore size distribution of 20–160 μm and pore wall thickness of 3–35 μm	Acellular scaffold	Adult male Wistar rats weighing 250 g, the number of animals used was not stated; Surgical brain trauma was induced in rats < 24 hours of implant duration	No result for cell survival (degradation rate too rapid) Toxic by-products reported. Microglial cells and microphages detected at the edges of scaffolds. Improved cell survival	No functional recovery testing
Elias and Spector, 2012 ⁴⁰	Type 1 collagen based porous scaffold	Freeze-drying and dehydrothermal crosslinking yielded pore size distribution of 100–150 μm	Bromodeoxyuridine-labelled adult hippocampal neural progenitors	Male Sprague-Dawley rats, 12 animals tested using the penetrating brain injury model 4 weeks implant duration	Progenitors differentiated into astrocytes, endothelial cells, and oligodendrocytes. Angiogenesis noted	No functional recovery testing
Huang and co-workers, 2012 ³⁸	Collagen-glycosaminoglycan copolymer based porous matrix	Freeze-drying yielded 95% pore volume fraction and average pore diameter of $140 \pm 20 \mu\text{m}$.	Acellular scaffold	Adult male Sprague-Dawley rats weighing between 300–350 g; 30 animals (10 per group); surgical brain trauma was induced 4 week implant duration	Proliferation, differentiation and migration of endogenous neural precursor cells Increase in differentiated cells over time (DCX+) Decrease in number of inflammatory cells (ED1+) at the lesion boundary zone over time	Lowered neurological deficit (neurological severity scores, mNSS, test) Improved sensorimotor recovery (tactile adhesive-removal test)
Cheng and co-workers, 2013 ⁴¹	Laminin derived IKVAV linked to self-assembling peptide RADA16 based hydrogel	Self-assembly yielded functionalized self-assembling peptide hydrogel	Encapsulated rat neural stem cells (HCN-A94-2)	Sprague-Dawley rats, 24 rats used, underwent rat brain surgery model 6 week implant duration	Increased survival of embedded neural stem cells (NSC) Differentiation of embedded NSCs in functionalized scaffold (increased b-tubulin positive cells) Lesion sites expressed neuronal markers such as MAP2 and NF-H Neurite outgrowth and extension GFAP positive astrocytes were significantly reduced at the boundary of damaged cavity	No functional recovery testing

Table 1 (Contd.)

Reference and publication date of work	Scaffold system/type	Scaffold fabrication methods and resultant architectural features	Cell-seeding/encapsulation	<i>In vivo</i> animal model and number of animals tested and duration of implant analysis	Effect of scaffold at lesion site	Functional behavioural assessment and improvement
Guan and co-workers, 2013 ¹⁶	Collagen (type 1) based porous gelatinous scaffold	Porous gelatinous scaffold, pore sizes ranged 100-300 µm	Human bone marrow-derived mesenchymal stem cell (hMSC)	Adult male Sprague-Dawley rats, 42 animals tested, weighing 230–250 g, were subjected to a controlled cortical impact injury 4 week implant duration	Improved cell survival	Improved neurological recovery (lower neurological severity score (mNSS))
Álvarez and co-workers, 2014 ⁴²	Poly-L/DL lactic acid (PLA70/30) nanofibers	Electrospinning yielded radially aligned electrospun nanofibers	Acellular scaffold	Mice, four days old mice 15 month implant duration	Improved neurite outgrowth Neural tissue regeneration with: – Functional laminar organization – Postsynaptic density protein 95 (PSD95) and synaptosomal-associated protein 25 (SNAP25) indicative of functional synaptic machinery in the regenerated tissue Sustained neurogenesis Sustained cell survival Differentiation and integration Functional vascularization (angiogenesis noted) Integration of neurons with brain circuitry establishing synaptic contacts Increased proliferation of NSC	Improved spatial learning (Morris water maze) No functional recovery testing
Hsieh and co-workers, 2015 ⁴³	Polyurethane based hydrogel	3D bioprinting (fused deposition manufacturing) yielding a thermoresponsive hydrogel	Murine neural stem cells	Adult AB (wild type) zebrafish 8 day implant duration		Improved motor response (spontaneous coiling)
Martínez-Ramos and co-workers, 2015 ⁴⁴	Ethyl acrylate (OEA) and hydroxyl ethyl acrylate (HEA) copolymer based channeled scaffold	Porogen-template leaching yielded polyacrylate copolymer channeled scaffold	Acellular scaffold	Wister female rats, weighing between 220–250 g, underwent cryogenic traumatic brain injury (cryolesion) 8 week implant analysis	Infiltration and extension of neurons inside scaffold (Tuj-1) Low neuronal survival in surrounding zone Neo-angiogenesis (new blood vessels) noted – Proliferation of endothelial cells – Endothelial cells formed lumen and organized into tubules	Improved CNS recovery (hatching rate) No functional recovery testing

Table 1 (Contd.)

Reference and publication date of work	Scaffold system/type	Scaffold fabrication methods and resultant architectural features	Cell-seeding/encapsulation	<i>In vivo</i> animal model and number of animals tested and duration of implant analysis	Effect of scaffold at lesion site	Functional behavioural assessment and improvement
Rivet and co-workers, 2015 ⁴⁵	Fibronectin/poly-L-lactide and agarose/methylcellulose based electrospun fiber and hydrogel hybrid	A combination of electrospinning (fiber component) Thermogelation (hydrogel component)	Acellular scaffold	Male Wister rat, 36 animals, underwent surgical brain trauma >60 day implant duration	Reduced inflammatory response and glial scarring Formation of a loosely defined glial boundary Extensive infiltration of astrocytes (GFAP positive) and macrophages/microglia (ED1) at tissue-scaffold interface	No functional recovery testing
Zhang and co-workers, 2016 ²¹	Modified RADA16mix (RADA16-RGD and RADA16-IKVAV) based hydrogel	Yielded electrospun fiber dispersed in hydrogel Self-assembly peptide nanofibrous scaffold	Acellular scaffold	Adult C57BL/6 mice 2–3 months old, weighing between 22–28 g, subjected to rat intracerebral haemorrhage 6 week implant duration	Increased neuronal survival (increased NeuN-positive cells around injured area) Decreased cell death GFAP-positive and Iba1-positive cell found inside scaffold No NeuN positive cells infiltrated inside scaffold Nerve fibers present inside scaffold Decreased inflammatory response (lowered IBA-1 and CD11b) around scaffold Highly transparent sheet <i>in vivo</i>	Improved motor coordination skill recovery (rotarod test) Faster recovery of tension force of forelimb (grip strength test) improvement noted with catwalk analysis
Wang and co-workers, 2017 ⁴⁶	Polydimethylsiloxane, multiwalled carbon nanotube sheet	Printing transfer method yielded transparent multiwalled carbon nanotubes (MWNTs) coated polydimethylsiloxane sheet Gelation method yielded injectable, porous hydrogel	Acellular scaffold	Sprague-Dawley rats, 20 weeks old Duration of implant not stated		No functional recovery testing
Zhang and co-workers, 2018 ⁴⁷	Sodium alginate and hyaluronic acid based hydrogel		Human umbilical cord mesenchymal stem cells (hUC-MSC)	Sprague-Dawley rats weighing 230–250 g), subjected to Feeney's weight-drop method 28 day implant duration	Improved cell viability ratio and proliferation	Functional recovery of nerve function evaluated using the modified neurologic severity score (mNSS) (motor, sensory, balance, and reflexes in rats) • Improved recovery of exercise, learning and memory abilities Learning ability and memory recall was investigated using a Morris water maze system

Table 1 (Contd.)

Reference and publication date of work	Scaffold system/type	Scaffold fabrication methods and resultant architectural features	Cell-seeding/encapsulation	<i>In vivo</i> animal model and number of animals tested and duration of implant analysis	Effect of scaffold at lesion site	Functional behavioural assessment and improvement
Motalleb and co-workers, 2018 ⁴⁸	Peptide amphiphile and tenascin-C derived peptide sequence based hydrogel	Self-assembly yielded and injectable <i>in situ</i> gelling hydrogel	Acellular scaffold	Male Sprague-Dawley rats, 6–7 weeks old 7 days	Redirected endogenous neuroblasts towards injury site Did not elicit a neuroinflammatory response	No functional recovery testing

Abbreviations and symbols: RADA16 = AcN-RADARADARADA-CONH₂,⁴¹ IKVAV = isoleucine-lysine-valine-alanine-valine,⁴¹ MAP-2 = Microtubule associated protein 2; NF-H = Anti-neurofilament H antibody, GFAP = Glial fibrillary acidic protein; NeuN = Hexaribonucleotide Binding Protein-3, NSC = Neural stem cell; Tuj-1 = anti-neural class IIIB-tubulin; DCX = Neuronal migration protein doublecortin, ED1 = mouse monoclonal antibody; CD11b = an alpha subunit of Mac-1, IBA-1 = ionized calcium-binding adapter molecule 1.

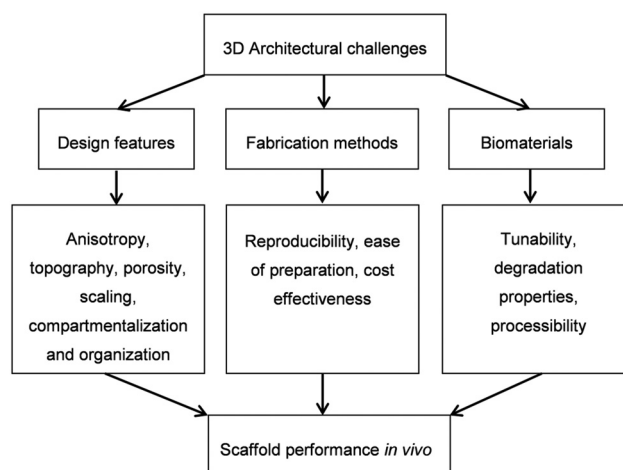


Fig. 2 Schematic outline of the review paper for the evaluation of 3D scaffolds for neural regeneration and repair.

infiltration) as an additional measurement tool to compare these 3D scaffold architectures. What is evident from these scaffold designs is the multimodal design approach, indicative of the efforts of this period (2011 to date). Three dimensional architectures enable and accommodate multimodality/complexity of design.

Incorporating multiple modes of interaction enables the scaffold to accommodate the dynamic needs of the regeneration process from biological (*e.g.* cell replacement, delivery of growth factors) to architectural stimulation (*e.g.* protective 3D environment, nanofibers to direct axonal extension).

In terms of biomimicry, both solid and liquid systems can exhibit viscoelastic behaviour similar to brain tissue. This is important in tissue engineering as it allows the 3D scaffold to retain its shape during transplantation and when experiencing the stress/strain that neural tissue is normally exposed to. Furthermore, brain tissue is stable and exhibits non-linear viscoelastic behaviour, highlighting that it can retain its shape after exposure to stress.⁵⁸

A crucial consult point in the architectural design is the translational potential of the scaffold system. That is, the scaffold architecture needs to be robust and accommodating of the clinical conditions (*e.g.* lesion shape) and procedures (*e.g.* transplantation stress) involved with the implantation process. Hydrogel scaffold systems possess numerous desirable features, *i.e.* they may be functionalized with bioactive compounds, have a low stiffness, high water content and conform to the geometry of any lesion.^{4,27,43,59,60} From a translational potential perspective, it is the ability of hydrogels to be formed *in situ*, which is advantageous, as the liquid can be delivered *via* syringe and needles allowing access to deeply located lesions in the brain while minimizing further disruption of the highly complex surrounding tissue network at the implantation site (*i.e.* implantation associated trauma). Alternatively, solid systems (*e.g.* nanofibers and nerve conduits) suffer from a translational perspective. They lack the fluidity of hydrogels which enables hydrogels to fill irregular

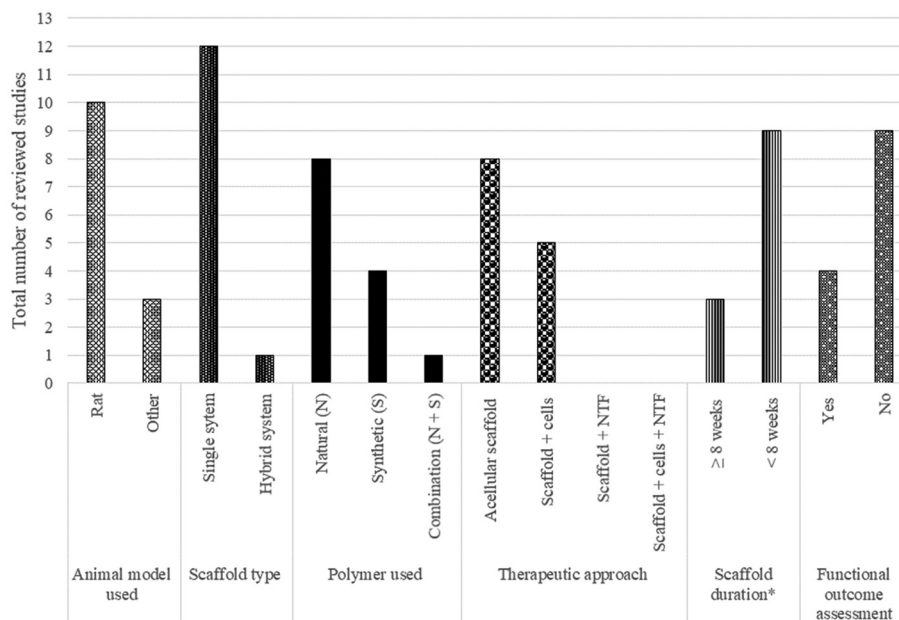


Fig. 3 Analysis of the reviewed 3D scaffold experimental studies reported in Table 1. Note: The total number of reviewed studies is 13. One of the reviewed studies⁴⁷ did not report on the duration of the implant in the animal model tested, therefore the “scaffold duration*” total is 12 instead of 13. NTF stands for neurotrophic factor.

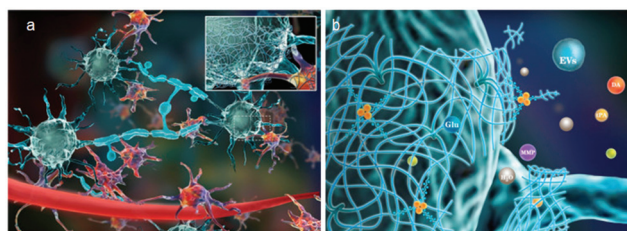


Fig. 4 Schematic depiction of the neural environment. (a) Spatial arrangement of the brain tissue compartments (cyan cells = neurons, purple cells = astrocytes, zoom-in subplot: ECM = a net-like cover attached to the surface of neural cells). (b) Enlarged view of the zoom-in subplot of the brain ECM in (a). The net-like cover = hyaluronic acid (HA) secreted by cells acting as a trestle for the ECM net-like structure. The penniform objects crossing the gaps between the HA chains = glycoproteins. Orange triple-circle = tenascin linking glycoproteins up. Multi-coloured single circles = components of interstitial fluid (ISF), e.g., water molecules (H_2O), extracellular vesicles (EVs), matrix metalloproteinase (MMP), glucose (Glu), dopamine (DA), and tissue plasminogen activators (tPAs). This figure is reused and reprinted from *Progress in Neurobiology*, Lei and co-workers.⁵² The brain interstitial system: anatomy, modelling, *in vivo* measurement, and applications, 2017, under the terms of the Creative Commons Attribution License (CC BY).

lesion geometries. Therefore, application of these scaffolds may result in further injury where tissue resection may be required to accommodate the 3D scaffold architecture. Alternatively, the 3D scaffold may need to be cut, shaped or resized to the shape of the lesion.^{37,39,44,57} Surprisingly, only a few scaffolds were inserted by the minimally invasive injection route,^{15,43,47,48,61,62} while the majority were surgically implanted *via* manual transfer (Table 1). This not only adds an

additional transplantation procedure, it also threatens the architectural integrity of the scaffold. This is a crucial point to consider as most 3D solid scaffolds (e.g. nanofibers) exhibit anisotropy. Elias and Spector⁴⁰ noted their pre-cut collagen scaffold failed to fill the lesion cavity; this was because the PBI lesion cavity had expanded to some extent by the time of implantation. Alterations in lesion size must thus be expected after a period of injury and considered in scaffold design, and could pose a limitation for solid scaffolds.

Cryogels are gaining interest as 3D scaffold systems for neural tissue engineering. Cryogels confer the best translational features of both liquid (e.g. hydrogels) and solid (e.g. nanofibers) scaffold systems for CNS application. They are both robust and accommodating by virtue of their shape memory properties.^{63–65} Firstly, cryogels are compressible and may therefore be delivered *via* the minimally invasive injection route. Secondly, cryogels can withstand deformation in the needle while maintaining their predefined architecture. Furthermore, cryogels can expand to their original shape post compression and injection. Ultimately, cryogels are potentially architecturally advantageous as they are both injectable (as hydrogels are) and architecturally predefined and robust like other solid systems (e.g. nanofibers).

Cells respond to the mechanical properties of their growth substrate.²⁵ The 3D scaffold system needs to be soft and allow for deformation, properties mimetic of neural tissue.²⁸ Soft matrices simulating the mechanical properties of neural tissue (elastic modulus <1 kPa) result in improved neural stem cell differentiation, cell viability and neurite extension, which is crucial in attaining reconnection of severed neurons.^{44,63} Unfortunately, this renders the scaffold vulnerable to trans-

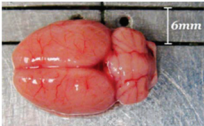
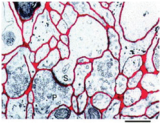
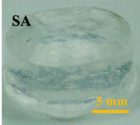
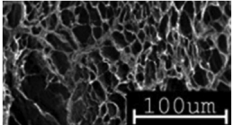
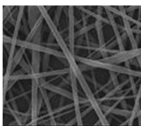
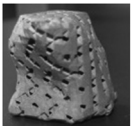
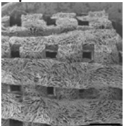
plantation stress. The architectural challenge associated with soft scaffold constructs is the risk of collapsing and damaging the CNS architecture and cellular network during implantation as observed by Jurga and co-workers.³⁹ Scaffold systems need to possess adequate mechanical properties to withstand the tension exposed to during regeneration across the lesion cavity.⁶⁶ In light of this, hybrid composite scaffold systems have been introduced. For example, Wu and co-workers⁶⁷ used an electrospun poly-L-(lactic acid) nanofibrous conduit to augment the outcome of a soft hydrogel for PNS regeneration. The hydrogel was injected inside the poly-L-(lactic acid) conduit, as the conduit possessed the required mechanical properties, which enabled it to maintain its bulk architectural morphology without collapsing before tissue could regenerate. Composite systems allow for better biomimicry of the target tissue by combining the advantages of both systems (solid and liquid) and overcoming their individual limitations.^{43,54} Composites allow for a more realistic mimicry of the ISS that is crucial for cell survival and differentiation. The 'liquid' components such as hydrogels mimic the interstitial fluid (ISF), which is a functional gel-like media. Whereas solid components such as nanofibers offers mechanical and architectural integrity conferred by the ECM.

2.2. Micro-architectural profiles of 3D scaffolds: compartmentalization, multi-scaling, mechanical anisotropy

2.2.1. Neuromimetic compartmentalization and multiscale organizational architectural features.

Brain tissue comprises structures of various sizes. For example, the neuronal cell body is approximately 10–50 μm in diameter, dendrites extend up to 500 μm , arteries and veins are 1–3 mm, and capillaries are approximately 4–8 μm .⁶⁸ This indicates that the ISS is multi-compartmental with various pore sizes to accommodate these various structures. As depicted in Table 2, the internal structure of the brain has previously been described as resembling foam.⁵⁰ The ISS (neural micro-environment) forms interconnected channels and cavities (spaces filled with cells) that are irregular in both size and shape. It is very interesting to note that the majority of the 3D scaffolds summarized in Table 1 are able to mimic this internal structure. The walls of the scaffolds presented in Table 2 can serve as the channels while the pores function as spaces to be occupied by cells. This structure is prominent in all the 3D systems (Table 1), regardless of whether conventional methods of fabrication (*e.g.* cryogelation, freeze-drying) or more advanced methods such as 3D printing were employed.

Table 2 Macro/micro architectural profiles of acellular biomaterial scaffolds in comparison to brain tissue

Tissue/ scaffold type	Macroarchitecture	Microarchitecture	Image reference
Brain tissue	 Rat brain after extraction	 Electron micrograph showing the extracellular space (red) outline from a region of a rat cortex. The scale bar measures 1 μm . "s" points to a prominent dendritic spine while "p" shows a presynaptic terminal	(a) ⁶⁹ (b) ⁵⁰
Hydrogel	 Photograph of sodium alginate (SA) hydrogel	 Scanning electron micrograph of a hydrogel.	(a, ⁴⁷ b ⁷⁰)
Nanofiber	 Photograph of nanofibrous mat	 Scanning electron micrograph of nanofibrous mat. The scale measure 5 μm .	(a,b) ⁷¹
3D printed scaffold	 3D printed scaffold with an external shape matching a brain lesion cavity	 Scanning electron microscopy image of a cross-sectional view of the 3D printed scaffold. The scale bar measures 1 mm.	(a,b) ⁷²

Symbols (a) and (b) are used to differentiate the corresponding image references in each row. Where (a) = image in macroarchitecture column, (b) = image in microarchitecture column.

Compartmentalization within the scaffold enables the development of various tissue structures (*e.g.* vascularization and neuronal networks) of various sizes and shape.²⁷ This can be achieved through the incorporation of features such as pores, grooves and channels. Interconnected pores are prevalent in 3D scaffold designs (Tables 1 and 2). Interconnected pores impart the properties of an open integrative system, that is, pores allow integration with the surrounding native tissue by facilitating cellular infiltration and tissue ingrowth which merges with the newly formed tissue within the scaffold, a property useful for neural tissue as reconnecting severed networks/circuits is key.

Compartmentalization allows the formation of concentration gradients and specialized niche-environments within the scaffold to aid in regeneration. These concentration gradients can be in the form of hormones, nutrients, undifferentiated stem cells.⁷³ Sart and co-workers⁷⁴ noted a higher survival of stem cells when they were prepared under conditions of hypoxic stress. This is called hypoxic conditioning. The injury site following traumatic brain injury is associated with oxidative stress, nutrient deprivation and lack of other pro-regenerative features (*e.g.* extracellular matrix) required for cell survival and tissue regeneration.⁷⁵ Antioxidants, survival signals and various genes related to cell migration, cell adhesion, and anti-inflammatory responses are regulated by the expression of HIF-1.⁷⁶ Hypoxia preconditioning (at sub-lethal hypoxic levels) is a strategy designed to mimic an aspect of this lesion environment to promote endogenous protective mechanisms in the cells to enhance survival in this environment.^{75,77} Hypoxic preconditioning directly induces HIF-1 upregulation which in turn increases brain-derived neurotrophic factor (BDNF), stromal-derived factor 1 (SDF-1), vascular endothelial growth factor (VEGF), erythropoietin (EPO), and many other genes that can stimulate neurogenesis, angiogenesis, vasodilatation, and increase cell survival.⁷⁶ This hypoxic preconditioning effect of cells may be achieved using 3D scaffolds by simply tailoring the diffusion properties of the scaffold through architecture (*e.g.* compartmentalization, pore size).⁷³ Therefore, 3D scaffolds can be used to replicate stages of concentration gradients (*e.g.* of oxygen and nutrients) that drive stem cell differentiation, and impart biological heterogeneity through controlled spatial organizational design.^{44,78} In the same note, cells in 3D constructs run the risk of suffering core degradation from lack of adequate diffusion (*e.g.* waste removal, oxygen diffusion) and vascularization. Interconnected pores enable the free flow of waste products and nutrients within the scaffolds, improving overall cell viability.⁷⁹ Vasculature formation is crucial to cell viability and porous structures facilitate the formation of vasculature.⁷⁹

2.2.2. Influencing neural cell fate through multi-scaling features. The native cell environment is inherently hierarchical/multiscaled (*i.e.* composed of nano- to micro- to macro-structures).^{80,81} Various scales may be used to mimic the ECM within the 3D scaffold design; nano-scale at 0.001–1 μm , micro-scale at 1–1000 μm and macro-scale 1000–100 000 μm .⁸²

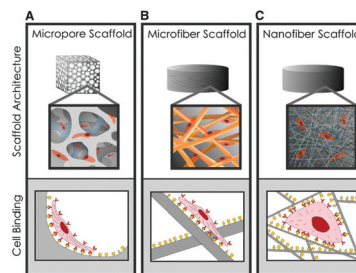


Fig. 5 Scaffold architecture affects cell binding and spreading. (a and b) Cells binding to scaffolds with microscale architectures flatten and spread. (c) Scaffolds with nanoscale architectures possess larger surface areas for protein absorption, presenting many more binding sites to cell membrane receptors. (a)–(c) are reused from Science, 310(5751), Stevens and George.⁸³ Exploring and engineering the cell surface interface, pp. 1135–1138, 2005, with permission from the American Association for the Advancement of Science.

Cell fate is influenced differently at these various scales 73,80,81 as depicted in Fig. 5.

Benefits and limitations exist for the various scales. Larger scaled scaffold features easily accommodate mass transport within the matrix and mimic 3D tissue structure.⁸² Limitations associated with these scales is the resultant 2D flat cell shapes and an imbalance in receptor attachment which is not ideal for 3D tissue formation (Fig. 5)⁸³ and cell interaction with their surrounding *via* nano-scale proteins, therefore a popular design route is down-scaling the physical dimensions to nano-scale.^{26,27,82,84} Nano-scale topography presents a higher surface area-volume ratio which enhances cell attachment (due to increased exposure of cells to binding sites), cell-scaffold interaction, differentiation and growth.^{25,74,81} Limitations of nanoscale features is that the size itself may limit cell infiltration, migration and overall mass transport within the scaffold, leading to 2D growth (a monolayer) on the surface of the scaffold instead of the desired 3D tissue.⁸⁴

Table 1 shows scaffolds with hierarchical organization. The above-mentioned limits and benefits of individual features provide some rationale for many of macro/micro/nano-architectural features observed with many scaffolds. A hierarchical architecture shows consideration for the average size of adult neural stem cells (approximately 20 μm) and overall tissue target features. Considering the average size of an adult neural stem cell, pore sizes ranging 20–160 μm reported in Table 1 proved to be optimal, small enough to support 3D cell–cell contacts for neural network structures to form and large enough to allow mass transport, and infiltration of neuro-endothelial cells for the formation of micro vessels post implantation.^{37,39,44} Jurga and co-workers³⁹ noted that cell differentiation and spatial compartmentalization could be modulated by pore size. That is, regions of undifferentiated and differentiated (nestin positive neural cells or GFAP positive glial cells) cells were co-existing within the same scaffold. It was hypothesized that it was because small pores (<20 μm) experienced limited penetration of morphogenic factors in

comparison to medium pores (80–120 μm).³⁹ Furthermore, possessing niche areas of undifferentiated cells within the scaffold mimics the native regions of neurogenesis in the brain such as the sub-ventricular zone (SVZ) and the dentate gyrus (DG). These regions of neurogenesis play a crucial role in regeneration and repair in conditions such as traumatic brain injury.^{79,82}

2.2.3. Mimicking neuromechanical anisotropy through architecture. Brain tissue is non-linear, has pronounced compression-tension asymmetry, and viscoelastic behaviour.^{58,84} These properties are an example of mechanical anisotropy, where the mechanical material behaviour is regionally dependent (mechanical properties vary when measured along different directions along the material).^{58,68,83–85} Anisotropy is a property of neural tissue imparted by the hierarchically ordered organization of its composites (*e.g.* cells, ECM and axonal extension).^{68,83,86} Mechanical anisotropy is largely linked to fibre alignment, in fact, one of the most distinctive features of nervous tissue (*e.g.* brain tissue), is the directionally aligned axonal tracts arising from development of axons in specific paths.^{68,87} Structure–activity relationships are important as cell behaviour is dependent on their environment; cells detect and respond to their substrate using mechano-transduction *via* nanoscale topography.^{73,88–90} Cells adhere to a substrate using focal adhesion mechanically linked to their cytoskeleton which transmits signals to the cell nucleus influencing a range of cell responses (*e.g.* cell shape) (Fig. 6).^{85,91–93} The environment experienced by the cells is anisotropic and mixed (*e.g.* interstitial fluid pressure and ECM strain).

As disorganized as the idea of anisotropy and heterogeneous material properties may seem, brain tissue is highly organized into functional neural circuits as demonstrated by diffusion tensor images (DTI) in the literature. Water moves more freely along axonal bundles *versus* perpendicular to those bundles due to fewer obstacles to obstruct movement along the fibres, therefore DTI uses anisotropy to detect fibre organization.⁹⁴ Anisotropy is therefore central in establishing the architectural arrangement correlative to desired scaffold function. Adding anisotropic properties renders the 3D scaffold more similar to native tissue as it too is composed of heterogeneous elements (*e.g.* cells, fibres, biomolecules, blood vessels) aligned in unique, specific ways. The efforts to replicate neural tissue anisotropy include stiffness, pores, fibre alignment, channels and other topographical features^{95–98} within the 3D scaffold structures as observed Table 1. Furthermore, the use of co-polymer biomaterials adds an element of anisotropy (*e.g.* different chemical bonding at different directions).

As popular as they may be, hydrogels generally have a poorly oriented architecture as they tend to lack the physical cellular-scale topographical features to facilitate cellular adhesion and migration, or promote directed axonal growth.^{99–102} This innate poor cell alignment and lack of guidance cues, renders it difficult to align cells to achieve regeneration similar to the native tissue.⁶⁶ Unlike most unmodified hydrogels, scaffold systems that demonstrate superior axonal guidance properties contain nano-scaled features (*e.g.* aligned fibres), laminar structures or channels.^{103,104}

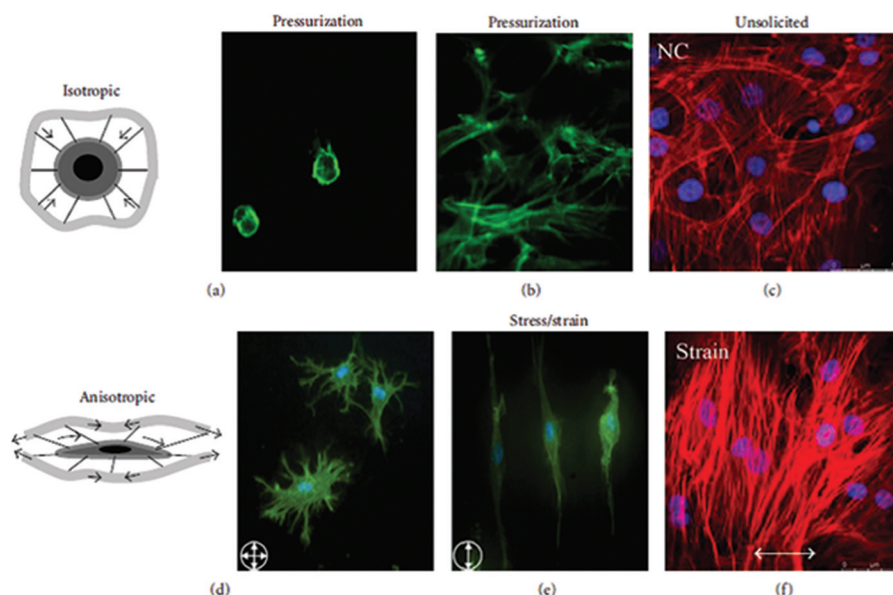


Fig. 6 Representation of substrate-directed cell fate (images not to scale). (a) Soft isotropic force resulting in a spherical cell shape and (b) spreading on stiffer substrate. (c) Random distribution of cells in isotropic conditions while (f) cell elongation is observed under anisotropic strain. (d) Cell protrusions in multiple directions is induced with biaxial strain whereas (e) uniaxial strain induces an elongated cell shape. This figure is reused and reprinted from Journal of Biomedicine and Biotechnology, Nava and co-workers.⁸⁵ Controlling self-renewal and differentiation of stem cells *via* mechanical cues, pp. 1–12, 2012, under the terms of the Creative Commons Attribution License.

Nanofibers present a nanoscale architecture which closely resembles the native ECM structure, with a large surface area to volume ratio, which allows the presentation of cellular-scale topographical cues across to foster guided neuronal migration and axonal extension across the lesion site.^{4,43,64} Post-implantation, Rivet and co-workers⁴⁵ noted the ability of astrocytes to not only locate the fibres within their 3D electrospun fibre and hydrogel hybrid scaffold, but also utilize their cues for migration into the hybrid matrix. This feature of guided growth and cell communication is a crucial advantage of a 3D scaffold considering its multidimensionality to ensure neurons extend towards reconnection instead of 'circling' within the planes of the 3D architecture. Electrospun poly (epsilon-caprolactone) nanofiber scaffolds were shown to influence the direction of neurite extension *via* perpendicular contact guidance.¹⁰⁵

In tissue engineering, anisotropic topography has proven effective in promoting directional neurite/axonal extension,¹⁰⁶ which mimics the *in vivo* environment where axons extend along the bands of Büngner created by glial cells. Bands of Büngner are longitudinal bands of Schwann cells that provide pathways to guide axon regeneration following lesion-induced Wallerian degeneration in lesioned nerves.^{107,108} Neurons and glial cells exhibited directed growth and bipolar shapes on aligned electrospun poly-L/D,L lactic acid (PLA70/30) nanofibers scaffolds, whereas randomly oriented, multipolar shapes were observed on random nanofibers (Fig. 7).⁴² Another approach is the use of aligned fibres which can guide cellular orientation and direction over distances.⁴ A 3D aligned poly(lactic-co-glycolic acid) (PLGA) fibre scaffold used to bridge a 3 mm spinal cord defect showed cells oriented along the direction of the scaffold and axons regenerating along the length of the scaffold.¹⁰⁹ A 3D scaffold of parallel fibres embedded in a matrix showed guided Schwann cell migration and axonal growth in the spinal cord.¹¹⁰ Fig. 7 depicts examples of topographical design features incorporated into 3D scaffolds.

2.3. Engineering neuromimetic architectures

Various methods have been employed to create porous 3D scaffolds in the recent years (2011 to date). Examples of these methods include cryogelation, freeze-drying, porogen template leaching, self-assembly peptides, electrospinning and 3D bio-printing (Table 1). Hydrogels are hydrated polymer networks whose dissolution in water is hindered by crosslinks.^{10,112} They are the most commonly fabricated scaffolds for neural tissue engineering. However, hydrogels have suffered the issue of insufficient vascularisation which could be improved with the addition of interconnected macropores.¹¹³ This could explain the frequency of pore producing methods (*e.g.* cryogelation, porogen-template leaching) observed in Table 1. The most common way of producing macroporous 3D hydrogels is by physically interrupting the hydrogel forming solution with pore forming units which are leached out post gelling leaving behind pores. The interconnections are formed in areas where the pore forming units have come into close proximity. Porogen leaching (also known as solution casting/particulate

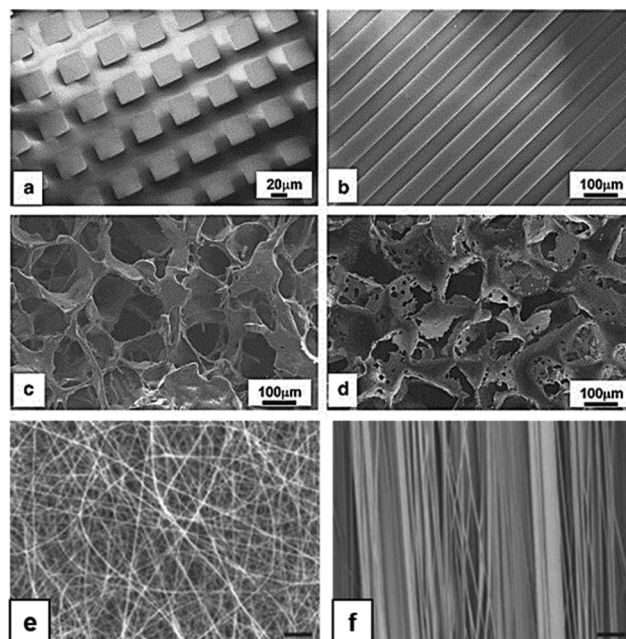


Fig. 7 Examples of topographical design features incorporated into 3D scaffolds represented by (a) squares, (b) grooves, (c, d) pores in polymeric scaffolds and (e) random and (f) aligned electrospun fibrous mats. The scale bars for (e) and (f) measure 10 µm. (a–d) is reused and reprinted from *Biotechnology Advances*, 30(1), Martino and co-workers.²⁵ Stem cell-biomaterial interactions for regenerative medicine, pp. 338–351, 2011, with permission from Elsevier. (e, f) is reused and reprinted from *Scientific Reports*, 6(1), Beliveau and co-workers.¹¹¹ Aligned nanotopography promotes a migratory state in glioblastoma multiforme tumor cells, p. 26143, 2014, under the terms of the Creative Commons Attribution 4.0 International License.

leaching), involves mixing a porogen (*e.g.* sodium chloride, sugar crystals) into a biodegradable polymer solution which is cast into a mould. Thereafter, the solvent is removed by lyophilisation while the porogen particles are leached out to obtain a porous structure. The limitation with this method is the poor interconnectivity, irregular pore geometry and difficulties associated with removing the porogens particles.¹¹⁴ Compared to porogen leaching, freeze-drying uses ice particulates as the porogen material, resulting in easier porogen removal. In this method, a polymer solvent solution and water emulsion is cooled, which in turn locks the liquid state structure. Thereafter the solvent and water are removed by freeze drying. Cryogelation is arguably one of the easiest methods of producing interconnected porous 3D scaffolds in comparison to other techniques (*e.g.* freeze-drying, porogen leaching).^{113,114} A polymeric solution is cooled below freezing point, resulting in crystallization of the solvent while the gel solution is maintained in a non-frozen liquid phase. The inherent interconnectivity of the pores within the 3D scaffold is a result of the solvent crystals growing towards each other and making contact with each other during crystallization.¹¹⁵ The porogen is removed by simple exposing the cryogel to temperatures above the solvent freezing point, which causes the solvent to melt, hydrating the polymeric gel network. The ability of these

methods to produce interconnected pores is highly desired to augment free flow of nutrients, waste products, cell migration and overall cell viability within the 3D scaffold. Self-assembly nanofiber scaffolds are alternative forms of hydrogels in which the fibrillary networks are formed either spontaneously in response to environmental changes such as pH, ionic strength and temperature.¹⁰ These networks then absorb water and form hydrogels. A large number of these fabrication methods are still highly conventional, and may in part play a role in limiting the success of the architectural design of scaffolds. It is important to note that these conventional methods were previously developed with the objective of drug delivery and not tissue engineering applications, however, they are being explored extensively for scaffolding in the field of tissue engineering.¹¹⁶ de Mulder and co-workers¹¹⁷ noted that one of the limitations of conventional fabrication methods is their production of isotropic scaffolds, an architecture that is different from the complex anisotropic architecture of native neural tissue. In addition, these conventional methods offer limited control over pore size, pore geometry, interconnectivity, and spatial distribution of the internal structure of a 3D scaffold.^{117,118} Cryogel³⁹ and freeze-dried scaffolds fabricated,^{37,39} resulted in a randomly distributed isotropic porous scaffolds, with a broad size range (20–160 μm).³⁹ The problem with such results is once again the randomly distributed porous architecture, which is alternate to the hierarchically ordered anisotropic property of neural tissue to facilitate directed neuronal growth.

Nanofiber scaffolds are characterised by their high surface area to volume ratio, high porosity and small (nano range) fibre diameter.¹⁰ Electrospinning is the most widely used method for fabrication of ECM-like 3D nano-fibrous scaffolds.^{4,90} Electrospinning uses electrostatic forces to form fine fibres with diameters in the nanometre range. However, electrospinning suffers from limited control, reproducibility and customization of architectural parameters. For example, it is difficult to produce equal fibre sizes with variable porosity as both properties are simultaneously modulated when altering the electrospinning parameters.¹¹⁹ Electrospinning is largely dependent on polymer solution properties, surface tension, conductivity and dielectric constant.⁹⁸ Some biomaterials are not suitable for scaffold fabrication *via* electrospinning due to high voltage or the high curing temperature and low solution viscosity.¹²⁰ This then limits the number of eligible materials for fabrication using this method. Additionally, a number of external factors further influence the resultant scaffold outcome *via* this approach, such as temperature, pressure and humidity.⁹⁸

Moreover, the majority of these conventional methods incorporate the use of solvents which often require an additional removal step as these solvents can be toxic to cells.^{100,109} Modifications to address the limitations of these conventional methods are under investigation. For example, an alternative would be the use of melt-electrospinning technique which does not require the use of toxic organic solvents needed for solution-electrospinning. Wu and co-workers⁶⁷

utilized a rotating metallic rod attached to a mandrel to achieve a poly-L-(lactic acid) conduit; this modified the electrospinning process by producing orientated/aligned fibres, which are more desirable than random fibre in terms of guiding the direction of neurite growth and cellular alignment.^{4,68,105}

More effort needs to be directed towards the fabrication of reproducible materials. Conventional methods have thus far made it difficult to not only adequately compare 3D scaffold designs, but to also elucidate successful architectural combinations as the details of the 3D internal structures (*e.g.* pore size, pore shape, fibre alignment) are not often adequately described (Table 1). Overall, conventional methods are limited in their ability to control the internal structure and precisely mimic the microtopography required for the formation of complex 3D architectures.^{102,106,109,110}

A progressive advance to accommodate the complexities of anisotropic 3D architectural scaffold fabrication is 3D printing, also known as the solid free form (SFF) technique or rapid prototyping technique.^{116,117,119} Three-dimensional printing techniques are computerized layer-by-layer fabrication methods capable of producing complex 3D architectures.^{101,109,117,116} The resultant 3D printed construct is informed by a computer aided design (CAD) which is often guided by anatomically correct imaging (employing, for example, magnetic resonance imaging (MRI) or computed tomography (CT) scans). Fig. 8 depicts an example of a typical 3D bioprinting process. The advantage of this approach is that the design can be rendered highly biomimetic, which increases the anatomical correctness of the construct. Additionally, due to application of a layer-by-layer manufacturing process, the resultant 3D printed scaffolds will display the sought after tissue property of anisotropy.¹²¹

A reproducible fabrication process is crucial for the commercial translation and potential clinical application of the scaffold. Three-dimensional printing techniques enable precise control of the internal architecture (*e.g.* spatial design, pore geometry) of the 3D scaffold with minimal material waste.^{101,109,117,120} The printing techniques are numerous; they may be liquid-based (*e.g.* stereolithography), solid-based (fused deposition modelling) or powder based. Fused deposition modelling is the most widely used of these techniques due to cost and reliability.¹²¹ Three-dimensional bioplotting is one of the latest major advancements in 3D printing technique in the field of tissue engineering.^{68,119} Processes such as bioplotting/bioprinting enhance the cell seeding process *via* co-fabrication, enabling the distribution of cells in desired locations. This removes the additional seeding stage required with conventional methods, and enables controlled spatial distribution of cells and nutrition.

Despite the potential for more controlled internal and external 3D scaffold architecture for customized patient specific scaffolds, much of the work on 3D printing in neural tissue engineering is still at the research and development (R&D) stage, evidenced by the low number of reports of *in vivo* work in the literature (Table 1). This can be attributed to the current

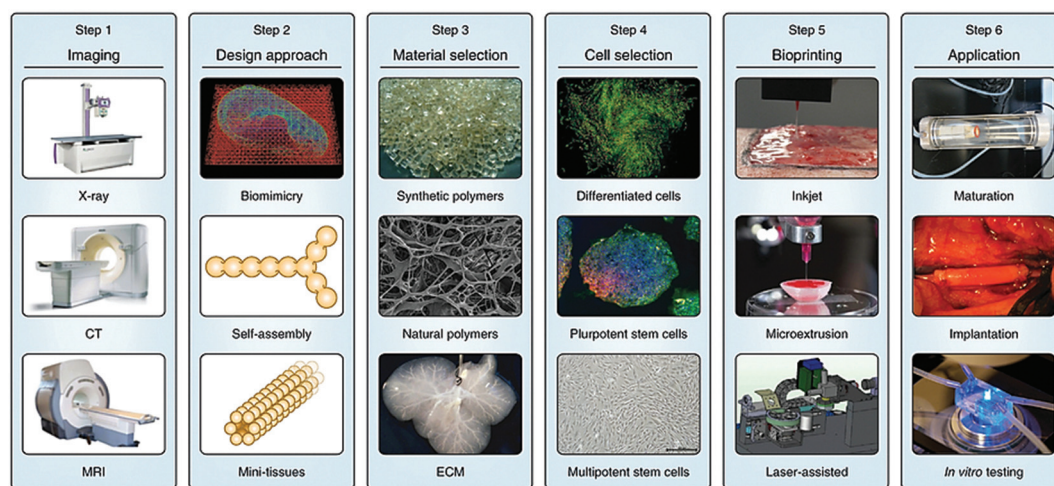


Fig. 8 An example of a typical 3D tissue bioprinting process. The process is simple; design of the printed tissue is guided by an image of the tissue (step 1) followed by choice of design approach (step 2). This is followed by input with suitable material and biological matter (steps 3 and 4). Feeding the instrument with the CAD and suitable materials yields a 3D printed tissue-mimetic construct (steps 5 and 6). This figure is reused and reprinted from *Nature Biotechnology*, **32**(8), Murphy and Atala.¹²⁷ 3D bioprinting of tissues and organs, pp. 773–785, 2014, with permission from Nature Publishing Group.

challenges facing 3D fabrication techniques with regards to neural tissue engineering.^{41,109}

A current challenge with 3D printing is the inability of the fabrication processes to accommodate the low modulus of neural tissue materials to produce complex stable architectures with good resolution.⁸⁰ A case in point is the 3D bioprinted neural stem cell-laden thermoresponsive biodegradable polyurethane hydrogel scaffold designed by Hsieh and co-workers.⁴³ A degree of structural collapse was observed with this scaffold during fabrication. The bioink possessed the low gel modulus desired to mimic soft neural tissue, however, the layer-by-layer deposition process failed to accommodate the softness of the material employed and resulted in structural collapse of the 3D scaffold, which compromised the integrity of the architecture.

Neural tissue possesses properties of soft tissue such as high elasticity, viscosity and flexibility coupled with large pores, which creates a problem when the fabrication process requires stacking the soft material (*e.g.* hydrogel) layer-by-layer through 3D printing.¹²² Hydrogels are most commonly used for brain tissue engineering, but their weak mechanical properties render it difficult for them to retain shape.¹⁰² Therefore, the conflict that arises is between the 3D printing process requirements and *in vivo* scaffold requirements. That is, for 3D printing the material needs to possess a low viscosity and gel rapidly to maintain the pre-designed geometry,^{118,123} whereas *in vivo* the material is required to be a soft matrix when in close proximity to neural tissue (<1 kPa elastic modulus) to augment neural stem cell differentiation and limit tissue stress/injury post implantation.

Material processability is a large hurdle which challenges 3D architectural scaffold fabrication resulting in a decline in the number of eligible materials when transitioning from con-

ventional fabrication methods to more controlled SFF methods. With regard to 3D printing, the rheological properties and degree of crosslinking largely influence the modulus of the material and thus printability.⁴³ For example, there is a limited range of biomaterials that can be processed by SFF techniques; they are limited to collagen, gelatin, fibrin, ceramics, thermoplastics or light curable composites.^{41,109,120,124} Furthermore, only a few of these are eligible for soft tissue application. Only a handful of commercial bioinks are available for 3D printing for soft tissue applications.¹²⁵ BioInk™ (RegenHu) is an example of a bioink containing fibrin/collagen for soft tissue application, however, this BioInk™ is for skeletal muscle tissues and not neural tissues.¹²⁶ To our knowledge, no bioinks currently exist for neural tissue. Moreover, none of the bioinks offer the option of functionalization,¹²⁵ limiting options for enhancing the regenerative properties of the scaffold.

Three-dimensional printing promises reproducible, personalized, clinically translatable, biomimetic 3D constructs and potentially *in situ/in vivo* bioprinting of scaffolds.^{127,128} Three-dimensional printing is far from maturity, especially with regards to neural tissue engineering, evidenced by the lack of bioinks and limited *in vivo* reports. However, 3D printing is undergoing rapid advances¹²² at a rate which needs to be met with concurrent improvements in resolution, printing speed, nozzle design, and most importantly, improvements in fabrication process compatibility with biocompatible materials and biomaterial advances (*e.g.* biomaterials for rapid crosslinking).^{122,127}

Ultimately, none of the current fabrication methods meet the standards required to create an optimal tissue-mimetic 3D architecture scaffold. Each are faced with their limitations, but strengths are also evident for each approach. For example, 3D

printing is the most promising technique for controlling micro/macroscale morphology to create the desired 3D physical structure.¹²⁹ With regards to neural tissue engineering, electrospinning remains the most promising technique for simulating the nanoarchitected fibrils found in the native ECM. While developments in precision printers and design of smaller needles to aid in increasing printing resolution are underway, perhaps a synergistic fabrication approach can be used in the meantime for neural tissue scaffolding. For example, Park and co-workers¹²⁹ demonstrated that combining 3D printing and electrospinning can yield a structure with improved nano-micro-macro architectural features. Even though 3D printing is the future, conventional methods are still leading in neural tissue engineering as evident in Table 1. Conventional fabrication methods should therefore, not be disregarded considering their large library of biomaterial options in comparison to the limited biomaterial options available for 3D printing.

2.4. Biomaterials for 3D neuro-scaffolding

As noted in the previous section, tissue engineering efforts are progressing towards more advanced fabrication methods such as 3D printing and sophisticated CAD. This could potentially drive more focus towards scrutinizing critical architectural design parameters in future work. Presently, a current major challenge may be the availability of suitable polymeric materials that not only produce biomimetic 3D scaffold architectures but that are also compatible with the processing requirements of current fabrication methods (e.g. electrospinning, 3D printing).

Therefore, this review found it worthwhile to attempt to define material properties for the fabrication of 3D scaffolds suitable for neural tissue application. Biomaterials are essential to the discussion and analysis of scaffold architecture. Not only do biomaterials enable the construction of these 3D architectures, they present an array of physicochemical properties to influence, or even enhance the scaffold capabilities to mimic anatomical structures. A wide variety of materials have been employed (Table 1), however the suitability of these materials is challenged by 3D scaffold architectural requirements. These biomaterial-specific architectural requirements include tunability (e.g. degradation rate, ability to conform to the desired shape, processability), maintenance of structural integrity, sterility, and biocompatibility.¹³⁰ The tunability of the material *in vivo* is very important for tissue regeneration in so far as providing a dynamic environment to accommodate the changes that occur with the tissue specific regeneration process (e.g. cell migration, tissue formation).

2.4.1. Biocompatibility. Biocompatibility of materials is pertinent to facilitate scaffold tissue integration.^{78,131} Biocompatibility requires that the material elicit no immune response and the surface chemistry allows for cell interaction (e.g. attachment sites). It is not surprising then that the majority (~61%) of the 3D scaffolds represented in the literature are composed of natural polymers (Fig. 3). Natural polymers such as collagen for example, display desirable properties

such as sites for cell adhesion; collagen forms one of the primary structural components of the native ECM and is present in abundance *in vivo*.^{64,132,133} Moreover, collagen scaffolds were noted to trigger brain-derived neurotrophic factor (BDNF, which regulates adult synapse plasticity in the CNS) and glial-cell derived neurotrophic factor (GDNF, which induces neuronal outgrowth release *in vivo*).³⁸ In contrast, synthetic polymers tend to lack the sites required for cell interaction. Synthetic polymers can be tuned to enhance their biocompatibility, through the addition of bioactive molecules (e.g. fibronectin, laminin, isoleucine-lysine-valine-alanine-valine (IKVAV)) and chemical alteration.³¹ By co-spinning fibronectin (natural polymer) and poly-L-lactide, Rivet and co-workers⁴⁵ tuned the cell adhesion properties of the poly-L-lactide nanofibers, resulting in improved cell infiltration into the matrix. Altering polymer chemical properties can also enhance the cell attraction of the construct. For example, hydrophilicity largely influences to cell attraction/affinity to the construct. Water soluble groups that contribute to hydrophilicity include amidic (–CONH–), ether, hydroxylic (–OH), sulphonic (–SO₃H), and carboxylic (–COOH) groups.^{134,135} The material and degradation products should be non-toxic ensuring no damage to newly formed/surrounding tissue.^{125,136} The self-assembly peptide (SAP) RADA-16-I has low pH (3.5) which is toxic to cells and tissues.²¹ Zhang and co-workers²¹ facilitated the formation of a hydrogel at a safer neutral pH by appending functional motifs, RGD and IKVAV to the SAP (Table 1). This markedly improved biocompatibility by lowering cytotoxicity of the scaffold while increasing cell adhesion (RGD) and neurite outgrowth (IKVAV). L-Lactate, the degradation product of electrospun PLA70/30 fibres, fuels angiogenesis and supports neuronal and neural cell metabolism.⁴²

2.4.2. Biodegradability. Biodegradability is an indicator of tunability; it is important that the material is biodegradable *in vivo*. From an architectural perspective, biodegradation of the scaffold material will enable the formation of predesigned structurally guided spaces upon which cells can deposit tissue-forming material. This also avoids unnecessary secondary injury from removal of the implant, which is ideal as scaffolds should serve as temporary surrogate support structure for neural tissue regeneration, and avoid unnecessary secondary injury on removal of the implant. In general, for a non-porous scaffold there are two forms of degradation: bulk and surface erosion. With bulk erosion, the rate of degradation is faster than the degradation of polymer bonds so degradation occurs throughout the structure, while surface erosion the degradation is limited to the polymer surface.^{121,126} Ideally the scaffold should exhibit surface erosion, as degradation from the outside in will maintain the shape and mechanical strength (avoiding collapse from mass loss) of the 3D construct architecture. However, with a porous material the presence of pores causes even a surface-degrading material to exhibit some degree of bulk erosion properties.

Consideration of the tissue target is crucial to the material properties of the scaffold construct. The implant site affects the degradation processes through the presence of water,

immune cells or enzymes. Most biodegradable polymers undergo hydrolytic degradation. The target tissue (brain tissue) has a very high water content (approximately 80% water).⁸ This high water content results in a high fluid flow that is able to replenish enzymes which may lead to increased degradation rates.⁸ The type of degradation and degradation rate affects the structural integrity of the scaffold *in vivo* and hence its performance. This requires careful consideration of the chemical properties of the polymer (*e.g.* hydrophilic: hydrophobic balance, crystallinity). For example, through consideration of functional groups, PLGA degradation rate can be tuned by altering the ratio of poly lactic acid (PLA) and poly glycolic acid (PGA) which alters its hydrophilic:hydrophobic balance. This is because the methyl side group of PLA renders it more hydrophobic than PGA.^{125–127,137} Therefore the more PGA present, the more hydrophilic, more water absorbing and subsequently faster degrading the PLGA.

Implant longevity *in vivo* ranged from <24 hours to 15 months (Table 1). An architectural challenge observed in some studies is the mismatch between the rate of tissue regrowth and scaffold degradation profiles, evidenced by the large range in duration of lifetime of implants *in vivo* in the various studies (Table 1). If the degradation rate is mismatched (*e.g.* too low or too high), there will be a resultant restriction in cell migration, proliferation and deficiencies within the scaffold, hindering neural tissue regeneration.^{37,78,109} An example of this is observed in a study by Rivet and co-workers,⁴⁵ where the 3D scaffold retained its shape for over 60 days. This was attributed to the material used being cellulose-based (methylcellulose) hydrogels which have a limited capacity for degradation because they are enzyme resistant and indigestible by the body.^{43,129,130} Therefore, the resilience (low degradation rate) of the methylcellulose used inhibited tissue regeneration efforts evidenced by the lack of reduction in lesion volume (Table 1). Another instance is when the degradation rate is too high. An example of this is observed in a study by Jurga and co-workers³⁹ where the 3D scaffold could only retain its shape for less than 24 hours. This was attributed to the material employed, being gelatin, which has a high degradation rate. The degree of crosslinking for this scaffold was also limited further by the amount of crosslinker used as the crosslinker, glutaraldehyde, can be toxic to cells. The combination of weak crosslinking and innately high degradation rate resulted in an accumulation of scaffold degradation components (gelatin and laminin) which produced a toxic, hypertonic, stressful environment for cells/tissue surrounding the implant site. Therefore, the fragility of the material and fabrication limitations, including the institution of a crosslinker, resulted in loss of mechanical and structural integrity of the implant premature to tissue formation. In both cases, the importance of biomaterial degradation rate (tunability) on 3D architecture performance is highlighted.

2.4.3. Biomaterial swelling to enhance architectural performance. Biomaterial swelling is the increase in volume that results from the migration of solvent molecules inside the

polymer network.^{135,138–141} This increase in volume in turn scales-up the 3D geometry of the scaffold architectural features.¹⁴² Biomaterial swelling is a considerable factor in scaffold design as it directly affects the porosity of the scaffold as well as its integration in the neural tissue. For example, if the pore walls are highly concentrated and thick, then additional water absorption within the polymeric pore walls could result in a marked reduction in pore size within the scaffold structure.¹⁴³ Biomaterial swelling may induce softening of a scaffold (*e.g.* hydrogel or cryogel) which in turn influences the mechanical properties and overall architectural integrity of the scaffold.

The reason that hydrogels are highly preferred materials for neural tissue engineering is due to their ability to swell. The swelling property is advantageous in terms of permeability, flexibility and reduced tensile strength.^{134,144} High water content results in an increased permeability/diffusion coefficient for enhanced mobility of solutes and metabolites.¹⁴⁵ The prerequisite for swelling is that the biomaterial has a deformable/flexible network. Free volume, thermal profiles (glass transition temperature) and crosslinking can aid in defining the suitability and flexibility of polymer networks. Free volume is the empty space/volume within a polymer material that is not occupied by the molecules,¹⁴⁶ not to be confused with pores. Free volume enables mobility of the polymer chains, a characteristic of amorphous materials, whereas more crystalline materials have a lower free volume and hence lower material flexibility. Free space within the polymer chain translates to increased flexibility of the polymer, and enhanced penetration of solvent molecules resulting in swelling.^{142,147} Therefore, semi-crystalline materials undergo less swelling than amorphous materials.

Swelling improves mimicry of the mechanical properties of neural tissue. Viscoelasticity is the property of neural tissue that enables it to exhibit both elastic and viscous characteristics when withstanding deformation. Penetration of solvent into the polymer network forces the polymer chains into an less entropically desired figuration; the chain in turn exerts a resistive/retractive force.^{134,141,144} This retractive force is what allows the biomaterials to withstand and recover from deformation, preventing architectural collapse.

Brain tissue is ~80% water⁸; the ability to swell allows the scaffold to maintain its architecture *in vivo* rather than dissolving. Crosslinking prevents polymers from undergoing hydrolysis in the fluid rich brain environment, and facilitates swelling.¹⁴⁸ The degree of crosslinking alters the swelling properties. Increasing the degree of crosslinking, decreases the extent of swelling by decreasing the free volume and increasing the glass transition temperature as the polymer backbones are drawn closer together and molecular mobility is reduced.^{139,141} This relationship between crosslinking and swelling is crucial from an architectural perspective as soft tissue materials are known for their poor mechanical properties. Therefore, careful balancing of the degree of crosslinking and swelling properties is paramount in processing and fabrication.

2.4.4. Processability. Materials need to be processed in order to produce the desired 3D scaffold architectural demands.¹¹⁶ Therefore, processability is a critical parameter with regard to 3D scaffolds. The general rule is that lower molecular weight materials are easier to process, however higher molecular weight materials possess better end use attributes.^{149–151} The disadvantage of semi-crystalline materials is that they are stiff, hence hard to process (*e.g.* melt or deform) into desired architectures. At the same time, their stiffness renders them more mechanically stable. In contrast, amorphous materials are more flexible, hence easier to process into desired architectures. At the same time, their flexibility confers mechanical instability, threatening structural collapse. Both these material features challenge both the fabrication methods and design phases of scaffolding. While synthetic polymers generally withstand harsh processing conditions such as heating (required with fused deposition and laser sintering technique), biopolymers tend to be limited to certain temperature and pH processing ranges.¹¹⁶

The composition of neural tissue is a crucial reference for consult when identifying desirable material properties. Various materials deform or respond differently to stimuli. Therefore the material properties of brain tissue are a result of the sum of the properties of these various materials. This may be one of the rationales behind the progression and success behind polycomposite biomaterials and modified polymers. One material may not necessarily be able to exert the range of properties to withstand processing required to produce a biomimetic 3D scaffold, however multiple components could be more successful. As presented in Table 1, the majority of the scaffolds in the recent years (from 2011 onward) utilize various copolymer combinations^{37,38,40,43,44,57} whereas only a few utilized a single polymer for fabrication.^{39,61} This trend of utilizing multiple polymers for fabrication highlights one of main avenues for addressing the major challenges of 3D scaffold design, which is the identification of a biocompatible, tunable, swelling biomaterial with sufficient biochemical and physico-mechanical properties to withstand both the fabrication process and the demands of a porous 3D architecture for neural regeneration.

2.5. *In vivo* tissue integration and performance of 3D scaffolds

The overall goal for 3D scaffolds is such that the architecture facilitates tissue integration with no anatomical or functional differences between the newly formed and native tissue. Physicochemical and design features such as porosity, tunability, and biomaterial swelling, as discussed, aid in tissue integration. For example, interconnected pores promote integration by facilitating penetration of native tissue into the scaffold, and newly formed tissue throughout the scaffold to encourage reconnection of circuits. The majority of scaffolds reported cellular infiltration, proliferation and enhanced cell survival qualified by qualitative markers such as microtubule-associated protein 2 (MAP-2) and NeuN inside and around the scaffold (Table 1). This can be attributed to the integrative

nature of the 3D scaffold architectures through interconnected pores facilitating cellular infiltration, and the 3D architecture enclosing and protecting the cells from the inhibitory environment at the lesion site. Biomaterial swelling enhances bonding properties and tissue integration *in vivo* as the brain is composed of approximately 80% water. Chemical modification of the biomaterial by introducing negative charges can enhance tissue integration *in vivo*, as the highly negatively charged glycosaminoglycans (GAGs) are noted to act as ECM crosslinkers and reservoirs for water and growth factors *in vivo*.¹⁴¹

Aside from the previously noted physicochemical and design properties, tissue integration can further be enhanced through the incorporation of bioactive materials (peptides, neurotrophic factors and other ECM components) and cells.^{40,43,132,144,148} Functionalization with bioactive groups aids in tissue integration by ‘connecting’ the scaffold to the native tissue. For example, proteins and peptides aid in cell adhesion and differentiation while neurotrophic factors can assist in tissue development.¹⁵² Surprisingly, none of the reviewed studies utilized growth factors, failing to take advantage of this feature to potentially enhance tissue integration (Fig. 3). Nonetheless, with the correct architectural composition, bioactive effects can still be achieved. For example, implantation of a collagen scaffold by Huang and co-workers³⁸ resulted in a sustained increase in tissue concentration of brain-derived neurotrophic factors (BDNF and GDNF); this scaffold thus rendered the administration of exogenous growth factors unnecessary. Moreover, the scaffold resulted in a lowered neurological deficit and improved sensorimotor recovery. The incorporation of a BDNF-mimetic non-peptide ligand to a PCL nanofiber scaffold showed improved tissue integration by increasing neuronal sprouting at the scaffold-tissue interface.⁶² The incorporation of a self-assembling peptide amphiphile carrying tenascin-c signal showed a 24-fold increase in the number of endogenous neuroblasts directed to the injury site.⁴⁸ The majority (~61%, Fig. 3) of the scaffolds evaluated were acellular^{37,38,43,44,57} while a few scaffolds were seeded with stem cells. Examples of cells investigated included bone marrow stromal cells, human amnion-derived multipotent progenitor cells, human umbilical cord mesenchymal stem cells, rat neural stem cells (HC-A94-2), and rat hippocampal cells.^{16,38,41,47} The purity of the cell lines vary, for example, bone marrow stromal cells have been modified to release BDNF.¹³ Functional recovery was reported at 14 days by Chen and co-workers⁶¹ the presence of human amnion-derived multipotent progenitor cells in the scaffold could have enhanced the regeneration potential of the scaffold through the release neurotrophic factors.

Very few studies reported on the infiltration of endothelial cells⁴⁰ or neoangiogenesis^{38,57} post-implantation. Most implants described lowered inflammation at the scaffold-tissue interface and variable degrees of glial scarring and regeneration were observed. Neuronal migration, proliferation, and differentiation of neural stem cells into 3D scaffolds was observed in the majority of the studies (Table 1). Neuronal/

axonal extension was observed by Martínez-Ramos and co-workers.⁴⁴

Tissue integration failure and immunorejection is associated with the degree of reactivity of astrocytes and microglia, and scar formation.^{9,143} Following injury, immune mediators are released within minutes and microglial presence can be observed for extended periods after injury (even up to 12 months).^{146,153} Reduced inflammatory response and glial scarring were observed, qualified by lowered inflammatory cells markers such as ED1+, GFAP, IBA-1, and CD11b (Table 1). These are indicators of the potential for successful tissue integration. Tissue integration is usually evaluated employing optical and histological analysis.^{144,150} Optical analysis provides an overall view of the injured area after implantation, which is useful for visualizing changes in lesion volume and 'sameness' of the regenerated and native tissue. Histological analysis enables a more detailed description of optical observations. Clinical translational challenges are observed with both optical and histological methods of analysis. Firstly, they provide qualitative results which can be misleading, rendering comparison of 3D scaffold performances in literature impractical.^{149,154} The misleading nature arises from the fact that 2D images may not provide an adequate representation of the 3D tissue formed.¹⁵⁵ Secondly, they require tissue resection to analyse tissue integration which would defeat the efforts of tissue regeneration. There is an opportunity to enhance clinical translational assessment through developing non-invasive imaging techniques such as chemical exchange saturation transfer magnetic resonance imaging (CEST MRI), which can monitor cell viability and compositional changes of the scaffold *in vivo*.¹⁵⁶ There needs to be a development and application of more quantitative methods of evaluation such as MRI *T2* spin-spin relaxation index¹⁵² to obtain improved correlation of the homogeneity of the tissue-scaffold interface/tissue integration. MRI *T2* spin-spin relaxation is a sensitive and quantifiable indicator of the proportion of bound to free water in biological tissues.¹⁵⁷ *T2* relaxation time is particularly sensitive to the degree of water compartmentalization due to the loss of magnetization of water to surrounding molecules.¹⁵⁷ The higher the degree of compartmentalised water the lower the *T2* relaxation time.^{157,158} Tissue damage induces changes that result in variations in free water fraction. Therefore, *T2* relaxation time deviations can be used to quantify these changes in tissue water content correlating it to tissue damage (e.g. toxicity of scaffold) *in vivo*. Luciano and co-workers¹⁵⁹ presented a protocol for brain sectioning in efforts to apply 3D printing technology in linking post-mortem MRI and histological observations more accurately, which may potentially aid in elucidating key pathological markers. Potentially this technology can further be developed to be applied on living tissue to assist in more closely monitoring and deciphering effects the 3D implanted scaffolds have at the injury site.

2.5.1. Translational potential of 3D scaffolds with *in vivo* experimental models of TBI. Functional recovery is an additional approach to assess the efficacy and success of

scaffold treatment. To date recovery of nervous tissue scaffolds has been incomplete and unsatisfactory.³⁸ Outcomes were reported as successful for ~30% of reviewed studies (Table 1). Examples of tests conducted were the neurological severity scores (mNSS) test,³⁸ rotarod test,²¹ and grip strength test.²¹ Functional outcomes were observed at periods ranging from 2 weeks to over 8 weeks.^{40,43,45,61} In future, efforts should be directed to establish the efficacy of the scaffolds at a functional level in animal models.

Study controls, animal models assigned, and methods of analysis require strict consult when evaluating the translational clinical potential of the scaffold and interpretation of regeneration results. In some instances the choice of controls was poor. For example Jurga and co-workers³⁹ utilized hippocampal slices for comparison with implanted brain lesion results. Scaffold performance and potential is almost exclusively limited to rat models (Table 1). Larger animal models were popular until the 1980s, however ethical concerns led to a shifts towards the use of rodents.¹⁶⁰ Although rodents have yielded mounting experimental results, there is yet to be any reported successful clinical translation.¹⁶⁰ The translational success of 3D scaffolds may be related to the innate differences in the biomechanical and physiological responses of rodent and human tissue, specifically neural tissue, to injury.^{159,160} Unlike humans, rats have a strong regenerative potential and heal at a faster rate.¹⁶¹ Rodents may not be able to capture the complexity and the response of human tissue to injury due to anatomical differences (rodent brains are lissencephalic whereas human brains are gyrencephalic).¹⁶⁰ In addition, rodents are small animals, the 'critical nerve gap' (the nerve gap at which regeneration cannot occur without the aid of bridging/nerve graft) is 1.5 cm for rats, while for humans it is 4 cm.¹⁶¹ The lesion gaps might be too small to translate to human application. Going forward, larger animal models need to be considered to closer replicate humans where the lesion sizes can be larger. Adding variety to the choice of animal models could avoid skewed species specific results that currently exist in literature. However, the ethical implications of this and interspecies-related differences that hinder the effective translation of results from animal models to humans could also witness a shift to alternatives, such as 3D-printed models (artificial tissue organs fabricated using 3D printing technologies¹⁶²).

Translational success may be augmented by establishing a clear definition of the injury.¹⁶³ Unfortunately, current traumatic brain injury models exhibit overlapping pathological outcomes.¹⁶⁴ This is further complicated by the innate heterogeneity of the traumatic brain injury pathophysiology.^{159,160,163,165} Neural tissue engineering efforts may benefit from standardizing tests and regeneration outcome measures (e.g. determining noteworthy result criteria). The National Institute of Neurological Disorders and Stroke (NINDS) is exerting effort towards establishing common data elements that can ultimately be employed to render the accurate comparison of results possible.¹⁶⁴ Multi-parametric advances in imaging

may assist in profiling the complex injury process and 3D scaffold-cell and 3D scaffold-tissue interaction.

3. Conclusion

The French architect, Robert le Ricolais (1894–1977) stated: ‘The art of structure is where to put the holes’. For this review, a suitable phrase would then be ‘the art of scaffolding is positioning of the pores and channels’. A critical and extensive review of 3D architectural challenges identified pores as one of the biggest challenges in 3D scaffolds for neural tissue engineering. This is evidenced by the pronounced influence of pores on numerous aspects of the architecture and resultant scaffold behaviour and performance. This is observed from the molecular level, where free volume affects biomaterial swelling and mechanical behaviour, to the micro/macro-level where size affects presentation of cell binding sites, cellular infiltration and tissue integration. With an increased focus on customized treatments, advances in design approaches (computer aided design) and fabrication methods (3D printing), it is crucial now more than ever for efforts to be increasingly directed towards defining such critical parameters for improved 3D scaffolds for neural tissue engineering.

From an architectural perspective, the overall scaling and design approach, material properties, and fabrication methods should be such that the resultant architecture meets the requirements summarized in Table 3. Scaling and design parameters ensure that the scaffold is large enough to enable proper handling for implantation, small enough to be implanted into precise lesion locations, and should enable the ingrowth of various nervous tissue composites (*e.g.* extracellular matrix, neurons and neuroglia, and blood vessels) and tissue formation of appropriate size to integrate with native tissue. Biomaterial parameters affect the constructs’ biomimetic profile. Fabrication parameters largely influence not only the reproducibility but affect the successful translation of the scaffold design and materials employed into functional 3D architectures.

3D neural scaffolds reviewed have shown experimental success in animal models, confirming the progressiveness of the field of tissue engineering. What is clear is that multimodality, customization and reproducibility of the 3D scaffold architecture are of major importance as there is progression towards commercial feasibility enticed by the success in *in vivo* animal models. However, more effort need to be directed towards scaffold architecture as this is the element that largely influences the success or failure of the relationship between fabrication techniques and biomaterials *in vivo*.

Another one of the major challenges identified in this review is the lack of clear and comparable data in the literature due to the versatility of the materials, fabrication methods, design approaches and injury models in this field of study. Having comparative data and controls can aid in elucidating successes and failures of these scaffolds. For example, testing of the polymeric material independently and in combination with hybrid scaffolds to determine their specific roles should be undertaken, as material composition may be important in determining which cell types are likely to migrate into the scaffold.¹⁶⁷ Outcome measures of scaffold performance need to be standardized to aid in comparison.

The challenges highlighted in this review have multiple causes including the complexity of biological systems, incomplete understanding of cell–ECM interactions, inability to best apply technical advances to synthetic methods, and lack of appropriate biomaterials. This is manifested by the current shortfall in optimizing the combination of design requirements, fabrication methods and biomaterials for 3D neural scaffolds. Architectural challenges can be addressed through the concurrent advancement of computational aided design, biomaterial advancement and fabrication techniques. This will lead to the development of a systemic understanding of architectural challenges by recognizing all elemental components from design, to processing, to product development for more effective customized features. This may simplify the complexity of architectural challenges by exposing structure–function relationships. Tissue-informed computational models could proposedly be integrated into the initial design approach of scaffolds going forward.

Table 3 Summary of critical architectural design parameters for 3D neural scaffolds observed in this review

Parameter		Requirement
Design	Macro-architecture	3D construct
		Pliant and soft while sufficiently robust (for clinical implantation stressors)
	Micro-architecture	Anisotropy and compartmentalization
	Porosity	Ideally ~90% ^{44,166}
	Pore size	Multi-scale
Biomaterial	Pore structure	Interconnected
	Mechanical	Viscoelastic (elasticity of brain tissue: 0.1–1 kPa (ref. 85))
	Stability and suitability	Swelling properties (to provide flexibility, withstand deformation to structure)
		Tunable (degradable material)
		Non-toxic degradation products
Fabrication method		Sterilizable
		Reproducible architecture

Conflicts of interest

The authors report no conflicts of interest.

Acknowledgements

The work reported herein was made possible through funding by the Claude Leon Foundation, the National Research Foundation (NRF) of South Africa and the South African Medical Research Council through its Division of Research Capacity Development under the National Health Scholarship Programme from funding received from the Public Health Enhancement Fund/National Department of Health. The content hereof is the sole responsibility of the authors and do not necessarily represent the official views of the SAMRC or the funders.

References

- 1 J. He, X. Wang, M. Spector and F. Cui, Scaffolds for central nervous system tissue engineering, *Front. Mater. Sci.*, 2012, **6**(1), 1–25.
- 2 Z. Su, W. Niu, M.-L. Liu, Y. Zou and C.-L. Zhang, In vivo conversion of astrocytes to neurons in the injured adult spinal cord, *Nat. Commun.*, 2014, **5**, 3338.
- 3 E. Tamariz and A. Varela-Echavarría, The discovery of the growth cone and its influence on the study of axon guidance, *Front. Neuroanat.*, 2015, **9**, 51.
- 4 L. Tian, M. P. Prabhakaran and S. Ramakrishna, Strategies for regeneration of components of nervous system: scaffolds, cells and biomolecules, *Regener. Biomater.*, 2015, **2**(1), 31–45.
- 5 S. Casarosa, Y. Bozzi and L. Conti, Neural stem cells: ready for therapeutic applications?, *Mol. Cell. Ther.*, 2014, **2**(1), 31.
- 6 S. Das, A. Carnicer-Lombarte, J. W. Fawcett and U. Bora, Bio-inspired nano tools for neuroscience, *Prog. Neurobiol.*, 2016, **142**, 1–22.
- 7 P. Hao, H. Duan, F. Hao, L. Chen, M. Sun, K. S. Fan, *et al.*, Neural repair by NT3-chitosan via enhancement of endogenous neurogenesis after adult focal aspiration brain injury, *Biomaterials*, 2017, **140**, 88–102.
- 8 E. Aurand, J. Wagner, C. Lanning and K. Bjugstad, Building Biocompatible Hydrogels for Tissue Engineering of the Brain and Spinal Cord, *J. Funct. Biomater.*, 2012, **3**(4), 839–863.
- 9 C. Davoust, B. Plas, A. Bédier, B. Demain, A.-S. Salabert, J. C. Sol, *et al.*, Regenerative potential of primary adult human neural stem cells on micropatterned bio-implants boosts motor recovery, *Stem Cell Res. Ther.*, 2017, **8**(1), 253.
- 10 J. T. S. Pettikiriachchi, C. L. Parish, M. S. Shoichet, J. S. Forsythe and D. R. Nisbet, Biomaterials for brain tissue engineering, *Aust. J. Chem.*, 2010, **63**, 1143–1154.
- Available from: [http://www.ecf.utoronto.ca/~molly/publications/biomaterials for brain tissue engineering.pdf](http://www.ecf.utoronto.ca/~molly/publications/biomaterials%20for%20brain%20tissue%20engineering.pdf) [cited 2017 Aug 26].
- 11 A. Sarnowska, A. Jablonska, M. Jurga, M. Dainiak, L. Strojek, K. Drela, *et al.*, Encapsulation of mesenchymal stem cells by bioscaffolds protects cell survival and attenuates neuroinflammatory reaction in injured brain tissue after transplantation, *Cell Transplant.*, 2013, **22**(Suppl. 1), 67–82.
- 12 A. Heile and T. Brinker, Clinical translation of stem cell therapy in traumatic brain injury: the potential of encapsulated mesenchymal cell biodelivery of glucagon-like peptide-1, *Dialogues Clin Neurosci.*, 2011, **13**(3), 279–286.
- 13 M. I. Günther, N. Weidner, R. Müller and A. Blesch, Cell-seeded alginate hydrogel scaffolds promote directed linear axonal regeneration in the injured rat spinal cord, *Acta Biomater.*, 2015, **27**, 140–150.
- 14 H. Ghuman and M. Modo, Biomaterial applications in neural therapy and repair, *Chin. Neurosurg. J.*, 2016, **2**(1), 34.
- 15 T.-Y. Wang, K. F. Bruggeman, J. A. Kauhausen, A. L. Rodriguez, D. R. Nisbet and C. L. Parish, Functionalized composite scaffolds improve the engraftment of transplanted dopaminergic progenitors in a mouse model of Parkinson's disease, *Biomaterials*, 2016, **74**, 89–98.
- 16 J. Guan, Z. Zhu, R. C. Zhao, Z. Xiao, C. Wu, Q. Han, *et al.*, Transplantation of human mesenchymal stem cells loaded on collagen scaffolds for the treatment of traumatic brain injury in rats, *Biomaterials*, 2013, **34**(24), 5937–5946.
- 17 G. J. R. Delcroix, P. C. Schiller, J. P. Benoit and C. N. Montero-Menei, Adult cell therapy for brain neuronal damages and the role of tissue engineering, *Biomaterials*, 2010, **31**(8), 2105–2120.
- 18 D. W. Hwang, Y. Jin, D. H. Lee, H. Y. Kim, H. N. Cho, H. J. Chung, *et al.*, In vivo bioluminescence imaging for prolonged survival of transplanted human neural stem cells using 3D biocompatible scaffold in corticectomized rat model, *PLoS One*, 2014, **9**(9), e105129.
- 19 Y. Liang, P. Walczak and J. W. M. Bulte, The survival of engrafted neural stem cells within hyaluronic acid hydrogels, *Biomaterials*, 2013, **34**(22), 5521–5529.
- 20 W. T. Daly, A. M. Knight, H. Wang, B. R. De, G. Giusti, M. Dadsetan, *et al.*, Comparison and characterization of multiple biomaterial conduits for peripheral nerve repair, *Biomaterials*, 2013, **34**, 8630–8639.
- 21 N. Zhang, L. He and W. Wu, Self-assembling peptide nanofibrous hydrogel as a promising strategy in nerve repair after traumatic injury in the nervous system, *Neural Regen. Res.*, 2016, **11**, 717–718.
- 22 B. Samer, B. Grant, J. Lin, K. Epameinondas, W. Robert and K. Petros, The surgical management of nerve gaps, *Ann. Plast. Surg.*, 2018, **80**(3), 252–261.
- 23 D. P. Kuffler, Promoting axon regeneration and neurological recovery following traumatic peripheral nerve injuries, *Int J Neurorehabilitation*, 2015, **2**(1), 1–17.

- 24 G. N. Panagopoulos, P. D. Megaloikonomos and A. F. Mavrogenis, The present and future for peripheral nerve regeneration, *Orthopedics*, 2016, **40**(1), e141–e156.
- 25 S. Martino, F. D. Angelo, J. M. Kenny, S. Martino, F. D. Angelo, I. Armentano, *et al.*, Stem cell-biomaterial interactions for regenerative medicine, *Biotechnol. Adv.*, 2011, **30**(1), 338–351.
- 26 Z. Kuihua, W. Chunyang, F. Cunyi and M. Xiumei, Aligned SF/P(LLA-CL)-blended nanofibers encapsulating nerve growth factor for peripheral nerve regeneration, *J. Biomed. Mater. Res., Part A*, 2014, **102**, 2680–2691.
- 27 D. Grinsell and C. P. Keating, Peripheral nerve reconstruction after injury: A review of clinical and experimental therapies, *BioMed Res. Int.*, 2014, **2014**, 13.
- 28 A. M. Hopkins, E. DeSimone, K. Chwaleka and D. Kaplan, 3D in vitro modeling of the central nervous system, *Prog. Neurobiol.*, 2015, 1–25.
- 29 T. Sedaghati and A. M. Seifalian, Nanotechnology and bio-functionalisation for peripheral nerve regeneration, *Neural Regener. Res.*, 2015, **10**(8), 1191–1194.
- 30 W. Zhu, J. K. George, V. J. Sorger and L. G. Zhang, 3D printing scaffold coupled with low level light therapy for neural tissue regeneration, *Biofabrication*, 2017, **9**, 1–10.
- 31 P. Sensharma, G. Madhumathi, R. D. Jayant and A. K. Jaiswal, Biomaterials and cells for neural tissue engineering: Current choices, *Mater. Sci. Eng., C*, 2017, **77**, 1302–1315.
- 32 B. J. Pfister, T. Gordon, J. R. Loverde, A. S. Kochar, S. E. Mackinnon, D. K. Cullen, *et al.*, Biomedical engineering strategies for peripheral nerve repair: Surgical applications, state of the art, and future challenges, *Crit. Rev. Biomed. Eng.*, 2011, **39**(2), 81–124.
- 33 S. Kehoe, X. F. Zhang and D. Boyd, FDA approved guidance conduits and wraps for peripheral nerve injury: A review of materials and efficacy, *Injury*, 2012, **43**(5), 553–572.
- 34 P. Kumar, Y. Choonara, M. Girish, N. Dinesh and V. Pillay, Multifunctional Therapeutic Delivery Strategies for Effective Neuro-Regeneration Following Traumatic Spinal Cord Injury, in *Current Pharmaceutical Design*, Bentham Science Publishers, 2015, pp. 1517–1528.
- 35 W. Zhu, F. Masood, J. O'Brien and L. G. Zhang, Highly aligned nanocomposite scaffolds by electrospinning and electrospraying for neural tissue regeneration. electrospraying for neural tissue regeneration, *Nanomedicine*, 2015, **11**, 693–704.
- 36 T. H. Tung, Clinical strategies to enhance nerve regeneration, *Neural Regener. Res.*, 2015, **10**(1), 22–24.
- 37 R. H. Cholas, H. P. Hsu and M. Spector, The reparative response to cross-linked collagen-based scaffolds in a rat spinal cord gap model, *Biomaterials*, 2012, **33**(7), 2050–2059.
- 38 K. F. Huang, W. C. Hsu, W. T. Chiu and J. Y. Wang, Functional improvement and neurogenesis after collagen-GAG matrix implantation into surgical brain trauma, *Biomaterials*, 2012, **33**(7), 2067–2075.
- 39 M. Jurga, M. B. Dainiak, A. Sarnowska, A. Jablonska, A. Tripathi, F. M. Plieva, *et al.*, The performance of laminin-containing cryogel scaffolds in neural tissue regeneration, *Biomaterials*, 2011, **32**(13), 3423–3434.
- 40 P. Z. Elias and M. Spector, Implantation of a collagen scaffold seeded with adult rat hippocampal progenitors in a rat model of penetrating brain injury, *J. Neurosci. Methods*, 2012, **209**(1), 199–211.
- 41 T. Y. Cheng, M. H. Chen, W. H. Chang, M. Y. Huang and T. W. Wang, Neural stem cells encapsulated in a functionalized self-assembling peptide hydrogel for brain tissue engineering, *Biomaterials*, 2013, **34**(8), 2005–2016.
- 42 Z. Álvarez, O. Castaño, A. A. Castells, M. A. Mateos-Timoneda, J. A. Planell, E. Engel, *et al.*, Neurogenesis and vascularization of the damaged brain using a lactate-releasing biomimetic scaffold, *Biomaterials*, 2014, **35**(17), 4769–4781.
- 43 F. Y. Hsieh, H. H. Lin and H. Shui, 3D bioprinting of neural stem cell-laden thermoresponsive biodegradable polyurethane hydrogel and potential in central nervous system repair, *Biomaterials*, 2015, **71**, 48–57.
- 44 C. Martínez-Ramos, U. Gómez-Pinedo, M. A. G. Esparza, J. M. Soria, J. A. Barcia and M. Monleón Pradas, Neural tissue regeneration in experimental brain injury model with channeled scaffolds of acrylate copolymers, *Neurosci. Lett.*, 2015, **598**, 96–101.
- 45 C. J. Rivet, K. Zhou, R. J. Gilbert, D. I. Finkelstein and J. S. Forsythe, Cell infiltration into a 3D electrospun fiber and hydrogel hybrid scaffold implanted in the brain, *Biomatter*, 2015, **5**, e1005527.
- 46 C. Wang, S. Oh, H. A. Lee, J. Kang, K.-J. Jeong, S. W. Kang, *et al.*, In vivo feasibility test using transparent carbon nanotube-coated polydimethylsiloxane sheet at brain tissue and sciatic nerve, *J. Biomed. Mater. Res., Part A*, 2017, **105**(6), 1736–1745.
- 47 K. Zhang, Z. Shi, J. Zhou, Q. Xing, S. Ma, Q. Li, *et al.*, Potential application of an injectable hydrogel scaffold loaded with mesenchymal stem cells for treating traumatic brain injury, *J. Mater. Chem. B*, 2018, **6**(19), 2982–2992.
- 48 R. Motalleb, E. J. Berns, P. Patel, J. Gold, S. I. Stupp and H. G. Kuhn, In vivo migration of endogenous brain progenitor cells guided by an injectable peptide amphiphile biomaterial, *J. Tissue Eng. Regener. Med.*, 2018, **12**(4), e2123–e2133.
- 49 R. J. McMurtrey, Patterned and functionalized nano fiber scaffolds in three-dimensional hydrogel constructs enhance neurite outgrowth and directional control, *J. Neural. Eng.*, 2014, **11**, 15.
- 50 C. Nicholson and E. Syková, Extracellular space structure revealed by diffusion analysis, *Trends Neurosci.*, 1998, **21**(5), 207–221.
- 51 C. Shi, Y. Lei, H. Han, L. Zuo, J. Yan and Q. He, Transportation in the interstitial space of the brain can be regulated by neuronal excitation, *Sci. Rep.*, 2015, **5**, 1–11.

- 52 Y. Lei, H. Han, F. Yuan, A. Javeed and Y. Zhao, The brain interstitial system: Anatomy, modeling, in vivo measurement and applications, *Prog. Neurobiol.*, 2017, **157**, 230–246.
- 53 L. E. Bilston, *Neural Tissue Biomechanics Volume 3 of Studies in Mechanobiology, Tissue Engineering and Biomaterials. Illustrate*, ed. L. E. Bilston, Springer Science & Business Media, 2011, 288 p.
- 54 A. Wagner and W. Ehlers, A porous media model to describe the behaviour of brain tissue, *Proc. Appl. Math. Mech.*, 2008, **8**, 10201–10202.
- 55 X. Lin, J. Huang, Y. Shi and W. Liu, Tissue engineering and regenerative medicine in applied research: a year in review of 2014, *Tissue Eng., Part B*, 2015, **21**(2), 177–186.
- 56 C. Martínez-Ramos, A. Vallés-Lluch, J. M. G. Verdugo, J. L. G. Ribelles, J. A. Barcia, A. B. Orts, *et al.*, Channeled scaffolds implanted in adult rat brain, *J. Biomed. Mater. Res., Part A*, 2012, **100A**(12), 3276–3286.
- 57 L. A. Bosworth, L. Turner and S. H. Cartmell, State of the art composites comprising electrospun fibres coupled with hydrogels: a review, *Nanomedicine*, 2013, **9**(3), 322–335.
- 58 S. Budday, G. Sommer, C. Birkel, C. Langkammer, J. Haybaeck, J. Kohnert, *et al.*, Mechanical characterization of human brain tissue, *Acta Biomater.*, 2017, **48**, 319–340.
- 59 O. A. Carballo-Molina and I. Velasco, Hydrogels as scaffolds and delivery systems to enhance axonal regeneration after injuries, *Front. Cell. Neurosci.*, 2015, **9**, 13.
- 60 L.S. Moreira Teixeira, J. Feijen, C. A. van Blitterswijk, P. J. Dijkstra and M. Karperien, Enzyme-catalyzed cross-linkable hydrogels: Emerging strategies for tissue engineering, *Biomaterials*, 2012, **33**(5), 1281–1290.
- 61 Z. Chen, X. C. M. Lu, D. A. Shear, J. R. Dave, A. R. Davis, C. A. Evangelista, *et al.*, Synergism of human amnion-derived multipotent progenitor (AMP) cells and a collagen scaffold in promoting brain wound recovery: Pre-clinical studies in an experimental model of penetrating ballistic-like brain injury, *Brain Res.*, 2011, **1368**, 71–81.
- 62 D. Fon, K. Zhou, F. Ercole, F. Fehr, S. Marchesan, M. R. Minter, *et al.*, Nanofibrous scaffolds releasing a small molecule BDNF-mimetic for the re-direction of endogenous neuroblast migration in the brain, *Biomaterials*, 2014, **35**(9), 2692–2712.
- 63 T. C. Lim and M. Spector, Biomaterials for enhancing CNS repair, *Transl. Stroke Res.*, 2017, **8**(1), 57–64.
- 64 B. Newland, P. B. Welzel, H. Newland, C. Renneberg, P. Kolar, M. Tsurkan, *et al.*, Tackling cell transplantation anoikis: An injectable, shape memory cryogel microcarrier platform material for stem cell and neuronal cell growth, *Small*, 2015, **11**(38), 5047–5053.
- 65 A. Bédier, T. Braschler, O. Peric, G. Fantner, S. Mosser and P. Fraering, *et al.*, Injectable cryogels for neural tissue engineering applications, in *18th International Conference on Miniaturized Systems for Chemistry and Life Sciences*, San Antonio, Texas, United States of America, 2014, pp. 26–30.
- 66 S. Stratton, N. Shelke, K. Hoshino, S. Rudraiah and S. Kumbar, Bioactive polymeric scaffolds for tissue engineering, *Bioact. Mater.*, 2016, **1**(2), 93–108.
- 67 X. Wu, L. He, W. Li, H. Li, W.-M. Wong, S. Ramakrishna, *et al.*, Functional self-assembling peptide nanofiber hydrogel for peripheral nerve regeneration, *Regener. Biomater.*, 2016, 21–30.
- 68 M. A. Hemphill, S. Dauth, C. J. Yu, B. E. Dabiri and K. K. Parker, Traumatic brain injury and the neuronal microenvironment: A potential role for neuropathological mechanotransduction, *Neuron*, 2015, **85**(6), 1177–1192.
- 69 H. W. Haslach Jr., J. M. Gipple and L. N. Leahy, Influence of high deformation rate, brain region, transverse compression, and specimen size on rat brain shear stress morphology and magnitude, *J. Mech. Behav. Biomed. Mater.*, 2017, **68**, 88–102.
- 70 A. E. Forte, S. Galvan, F. Manieri, F. Rodriguez y Baena and D. Dini, A composite hydrogel for brain tissue phantoms, *Mater. Des.*, 2016, **112**, 227–238.
- 71 R. Ramachandran, V. R. Junnuthula, G. S. Gowd, A. Ashokan, J. Thomas, R. Peethambaran, *et al.*, Theranostic 3-Dimensional nano brain-implant for prolonged and localized treatment of recurrent glioma, *Sci. Rep.*, 2017, **7**, 43271.
- 72 F. Fu, Z. Qin, C. Xu, X.-Y. Chen, R.-X. Li, L.-N. Wang, *et al.*, Magnetic resonance imaging-three-dimensional printing technology fabricates customized scaffolds for brain tissue engineering, *Neural Regener. Res.*, 2017, **12**(4), 614–622. Available from: <http://www.ncbi.nlm.nih.gov/pubmed/28553343> [cited 2018 Mar 28].
- 73 R. J. McMurtrey, Multi-compartmental biomaterial scaffolds for patterning neural tissue organoids in models of neurodevelopment and tissue regeneration, *J. Tissue Eng.*, 2016, **7**, 1–8.
- 74 S. Sart, T. Ma and Y. Li, Preconditioning stem cells for in vivo delivery, *BioRes. Open Access*, 2014, **3**(4), 137–149.
- 75 S. Liu, J. Zhou, X. Zhang, Y. Liu, J. Chen, B. Hu, *et al.*, Strategies to Optimize Adult Stem Cell Therapy for Tissue Regeneration, *Int. J. Mol. Sci.*, 2016, **17**(6), 982.
- 76 Z. Z. Wei, Y.-B. Zhu, J. Y. Zhang, M. R. McCrary, S. Wang, Y.-B. Zhang, *et al.*, Priming of the cells: Hypoxic preconditioning for stem cell therapy, *Chin. Med. J.*, 2017, **130**(19), 2361–2374.
- 77 S. Baillieux, S. Chacaroun, S. Doutreleau, O. Detante, J. Pépin and S. Verges, Hypoxic conditioning and the central nervous system: A new therapeutic opportunity for brain and spinal cord injuries?, *Exp. Biol. Med.*, 2017, **242**(11), 1198–1206.
- 78 J. Kumari and A. Kumar, Development of polymer based cryogel matrix for transportation and storage of mammalian cells, *Scientific*, 2017, **7**, 1–13.
- 79 E. C. Chan, S.-M. Kuo, A. M. Kong, W. A. Morrison, G. J. Dusing, G. M. Mitchell, *et al.*, Three dimensional collagen scaffold promotes intrinsic vascularisation for

- tissue engineering applications, *PLoS One*, 2016, **11**(2), e0149799.
- 80 X. Tu, L. Wang, J. Wei, B. Wang, Y. Tang, J. Shi, *et al.*, 3D printed PEGDA microstructures for gelatin scaffold integration and neuron differentiation, *Microelectron. Eng.*, 2016, **158**, 30–34.
 - 81 S. Lee, W. Zhu, L. Heyburn, M. Nowicki, B. Harris and L. G. Zhang, Development of novel 3-D printed scaffolds with core-shell nanoparticles for nerve regeneration, *IEEE Trans. Biomed. Eng.*, 2017, **64**(2), 408–418.
 - 82 J. Lee, M. J. Cuddihy and N. A. Kotov, Three-dimensional cell culture matrices: State of the art, *Tissue Eng., Part B*, 2008, **14**(1), 61–86.
 - 83 M. M. Stevens and J. H. George, Exploring and engineering the cell surface interface, *Science*, 2005, **310**(5751), 1135–1138.
 - 84 S. Srinivasan, R. Jayakumar, K. P. Chennazhi, E. J. Levorson, A. G. Mikos and S. V. Nair, Multiscale fibrous scaffolds in regenerative medicine, *Adv. Polym. Sci.*, 2012, **246**, 1–20.
 - 85 M. M. Nava, M. T. Raimondi and R. Pietrabissa, Controlling self-renewal and differentiation of stem cells via mechanical cues, *J. Biomed. Biotechnol.*, 2012, **2012**, 1–12.
 - 86 A. Quinones-Hinojosa, N. Sanai, M. Soriano-Navarro, O. Gonzalez-Perez, Z. Mirzadeh, S. Gil-Perotin, *et al.*, Cellular composition and cytoarchitecture of the adult human subventricular zone: A niche of neural stem cells, *J. Comp. Neurol.*, 2006, **494**(3), 415–434.
 - 87 D. Sun, The potential of endogenous neurogenesis for brain repair and regeneration following traumatic brain injury, *Neural Regener. Res.*, 2014, **9**(7), 688–692.
 - 88 G. Franceschini, D. Bigoni, P. Regitnig and G. A. Holzapfel, Brain tissue deforms similarly to filled elastomers and follows consolidation theory, *J. Mech. Phys. Solids*, 2006, **54**, 2592–2620.
 - 89 L. Bilston, The influence of microstructure on neural tissue mechanics, in *Structure-Based Mechanics of Tissues and Organs. Illustrate*, ed. G. S. Kassab and M. S. Sacks, Springer, Boston, 2016, pp. 1–14.
 - 90 N. Oxman, E. Tsai and M. Firstenberg, Digital anisotropy: A variable elasticity rapid prototyping platform, *Virtual Phys. Prototyp.*, 2012, **7**(4), 261–274.
 - 91 A. S. Nathan, B. M. Baker, N. L. Nerurkar and R. L. Mauck, Mechano-topographic modulation of stem cell nuclear shape on nanofibrous scaffolds, *Acta Biomater.*, 2011, **7**(1), 57–66.
 - 92 A. J. Steward, S. D. Thorpe, T. Vinardell, C. T. Buckley, D. R. Wagner and D. J. Kelly, Cell-matrix interactions regulate mesenchymal stem cell response to hydrostatic pressure, *Acta Biomater.*, 2012, **8**(6), 2153–2159.
 - 93 Y. Huang, L. Zheng, X. Gong, X. Jia, W. Song, M. Liu, *et al.*, Effect of cyclic strain on cardiomyogenic differentiation of rat bone marrow derived mesenchymal stem cells, *PLoS One*, 2012, **7**(4), e34960.
 - 94 S. Mori and J. Zhang, Principles of diffusion tensor imaging and its applications to basic neuroscience research, *Neuron*, 2006, **51**, 527–539.
 - 95 T. Courtney, M. S. Sacks, J. Stankus, J. Guan and W. R. Wagner, Design and analysis of tissue engineering scaffolds that mimic soft tissue mechanical anisotropy, *Biomaterials*, 2006, **27**, 3631–3638.
 - 96 B. Xu, M. J. Chow and Y. Zhang, Experimental and modeling study of collagen scaffolds with the effects of cross-linking and fiber alignment, *Int. J. Biomater.*, 2011, **2011**, 172389.
 - 97 N. K. Mohtaram, J. Ko, C. King, L. Sun, N. Muller, M. B. Jun, *et al.*, Electrospun biomaterial scaffolds with varied topographies for neuronal differentiation of human-induced pluripotent stem cells, *J. Biomater. Res. Part A*, 2015, **103A**, 2591–2601.
 - 98 L. Krishna, K. Dhamodaran, C. Jayadev, K. Chatterjee, R. Shetty and S. S. Khora, Nanostructured scaffold as a determinant of stem cell fate, *Stem Cell Res. Ther.*, 2016, **7**(188), 1–12.
 - 99 L. Binan, C. Tendey, G. De Crescenzo, R. El Ayoubi, A. Ajji and M. Jolicoeur, Differentiation of neuronal stem cells into motor neurons using electrospun poly-L-lactic acid/gelatin scaffold, *Biomaterials*, 2014, **35**(2), 664–674.
 - 100 B. L. Du, X. Zeng, Y. H. Ma, B. Q. Lai, J. M. Wang, E. A. Ling, *et al.*, Graft of the gelatin sponge scaffold containing genetically-modified neural stem cells promotes cell differentiation, axon regeneration, and functional recovery in rat with spinal cord transection, *J. Biomed. Mater. Res., Part A*, 2015, **103**(4), 1533–1545.
 - 101 B. L. Du, C. G. Zeng, W. Zhang, D. P. Quan, E. A. Ling and Y. S. Zeng, A comparative study of gelatin sponge scaffolds and PLGA scaffolds transplanted to completely transected spinal cord of rat, *J. Biomed. Mater. Res., Part A*, 2014, **102**(6), 1715–1725.
 - 102 M. Büyüköz, E. Erdal and S. Alsoy Altinkaya, Nanofibrous gelatine scaffolds integrated with nerve growth factor-loaded alginate microspheres for brain tissue engineering, *J. Tissue Eng. Regener. Med.*, 2018, **12**(2), e707–e719.
 - 103 M. Georgiou, J. P. Golding, A. J. Loughlin, P. J. Kingham and J. B. Phillips, Engineered neural tissue with aligned, differentiated adipose-derived stem cells promotes peripheral nerve regeneration across a critical sized defect in rat sciatic nerve, *Biomaterials*, 2015, **37**, 242–251.
 - 104 G. R. Mitchell and A. Tojeira, Role of anisotropy in tissue engineering, *Procedia Eng.*, 2013, **59**, 117–125.
 - 105 D. R. Nisbet, A. E. Rodda, M. K. Horne, J. S. Forsythe and D. I. Finkelstein, Neurite infiltration and cellular response to electrospun polycaprolactone scaffolds implanted into the brain, *Biomaterials*, 2009, **30**(27), 4573–4580.
 - 106 W. Li, Q. Tang, A. D. Jadhav, A. Narang, W. X. Qian, P. Shi, *et al.*, Large-scale topographical screen for investigation of physical neural-guidance cues, *Sci. Rep.*, 2015, **2**(5), 1–8.
 - 107 V. T. Ribeiro-Resende, B. Koenig, S. Nichterwitz, S. Oberhoffner and B. Schlosshauer, Strategies for inducing the formation of bands of Büngner in peripheral nerve regeneration, *Biomaterials*, 2009, **30**(29), 5251–5259.

- 108 S. Rotshenker, Wallerian degeneration: the innate-immune response to traumatic nerve injury, *J. Neuroinflammation*, 2011, **8**, 109.
- 109 F. Zamani, M. Amani-Tehran, M. Latifi, M. A. Shokrgozar and A. Zaminy, Promotion of spinal cord axon regeneration by 3D nanofibrous core-sheath scaffolds, *J. Biomed. Mater. Res., Part A*, 2014, **102**(2), 506–513.
- 110 A. Kriebel, M. Rumman, M. Scheld, D. Hodde and G. Brook, Three-dimensional configuration of orientated fibers as guidance structures for cell migration and axonal growth, *J. Biomed. Mater. Res., Part B*, 2014, **11**(102), 356–365.
- 111 A. Beliveau, G. Thomas, J. Gong, Q. Wen and A. Jain, Aligned nanotopography promotes a migratory state in glioblastoma multiforme tumor cells, *Sci. Rep.*, 2016, **6**(1), 26143.
- 112 E. R. Aurand, K. J. Lampe and K. B. Bjugstad, Defining and designing polymers and hydrogels for neural tissue engineering, *Neurosci. Res.*, 2012, **72**(3), 199–213.
- 113 T. M. A. Henderson, K. Ladewig, D. N. Haylock, K. M. McLean and A. J. O'Connor, Cryogels for biomedical applications, *J. Mater. Chem. B*, 2013, **1**, 2682–2695.
- 114 S. A. Bencherif, T. M. Braschler and P. Renaud, Advances in the design of macroporous polymer scaffolds for potential applications in dentistry, *J. Periodontal Implant Sci.*, 2013, **43**(6), 251–261. Available from: <http://www.ncbi.nlm.nih.gov/pubmed/24455437>.
- 115 V. I. Lozinsky, F. M. Plieva, I. Y. Galaev and B. Mattiasson, The potential of polymeric cryogels in bioseparation, *Bioseparation*, 2001, **10**(4–5), 163–188.
- 116 R. C. Dutta, M. Dey, A. K. Dutta and B. Basu, Competent processing techniques for scaffolds in tissue engineering, *Biotechnol. Adv.*, 2017, **35**(2), 240–250.
- 117 E. L. W. de Mulder, P. Buma and G. Hannink, Anisotropic porous biodegradable scaffolds for musculoskeletal tissue engineering, *Materials (Basel)*, 2009, **2**(4), 1674–1696.
- 118 Q. L. Loh and C. Choong, Three-dimensional scaffolds for tissue engineering applications: role of porosity and pore size, *Tissue Eng., Part B*, 2013, **19**(6), 485–502.
- 119 A. L. Carlson, N. K. Bennett, N. L. Francis, A. Halikere, S. Clarke, J. C. Moore, *et al.*, Generation and transplantation of reprogrammed human neurons in the brain using 3D microtopographic scaffolds, *Nat. Commun.*, 2016, **7**, 1–10.
- 120 M. R. Badrossamay, K. Balachandran, A. K. Capulli, H. M. Golecki, A. Agarwal, J. A. Goss, *et al.*, Engineering hybrid polymer-protein super-aligned nanofibers via rotary jet spinning, *Biomaterials*, 2014, **35**(10), 3188–3197.
- 121 R. Zou, Y. Xia, S. Liu, P. Hu, W. Hou, Q. Hu, *et al.*, Isotropic and anisotropic elasticity and yielding of 3D printed material, *Composites, Part B*, 2016, **99**(99), 506–513.
- 122 J. I. Kim, T. I. Hwang, L. E. Aguilar, C. H. Park and C. S. Kim, A Controlled Design of Aligned and Random Nanofibers for 3D Bi-functionalized Nerve Conduits Fabricated via a Novel Electrospinning Set-up, *Sci. Rep.*, 2016, **6**, 23761.
- 123 T. Sudwilai, J. J. Ng, C. Boonkrai, N. Israsena, S. Chuangchote and P. Supaphol, Polypyrrole-coated electrospun poly(lactic acid) fibrous scaffold: effects of coating on electrical conductivity and neural cell growth, *J. Biomater. Sci., Polym. Ed.*, 2014, **25**(12), 1240–1252.
- 124 A. Mata, E. Jung, C. A. Boehm, A. J. Fleischman, G. F. Muschler and S. Roy, A three-dimensional scaffold with precise micro-architecture and surface, *Biomaterials*, 2009, **30**(27), 4610–4617.
- 125 M. Guvendiren, J. Molde, R. M. D. Soares and J. Kohn, Designing biomaterials for 3D printing, *ACS Biomater. Sci. Eng.*, 2016, **2**, 1679–1693.
- 126 B. K. Gu, D. J. Choi, S. J. Park, M. S. Kim, C. M. Kang and C.-H. Kim, 3-dimensional bioprinting for tissue engineering applications, *Biomater. Res.*, 2016, **20**(1), 12.
- 127 S. V. Murphy and A. Atala, 3D bioprinting of tissues and organs, *Nature Biotechnology*, Nature Publishing Group, 2014, vol. 32, pp. 773–785.
- 128 C. L. Ventola, Medical applications for 3D printing: current and projected uses, *P T*, 2014, **39**(10), 704–711.
- 129 S.-H. Park, U. H. Koh, D.-Y. Yang, N.-K. Lee and J. Hyun, Development of 3D Printed Biomimetic Scaffold for Tissue Engineering, *15th Int Conf Control Autom Syst (ICCAS. 2015 Oct;1958–60)*, 2015.
- 130 S. Snigdha, T. Sabu and E. Radhakrishnan, Polymer based tissue engineering strategies for neural regeneration, *Adv. Tissue Eng. Regen. Med.*, 2017, **2**(1), 1–6.
- 131 F. von Burkersroda, L. Schedl and A. Göpferich, Why degradable polymers undergo surface erosion or bulk erosion, *Biomaterials*, 2002, **23**, 4221–4231.
- 132 R. J. Mondschein, A. Kanitkar, C. B. Williams, S. S. Verbridge and T. E. Long, Polymer structure-property requirements for stereolithographic 3D printing of soft tissue engineering scaffolds, *Biomaterials*, Elsevier, 2017, vol. 140, pp. 170–188.
- 133 S. M. Willerth and S. E. Sakiyama-Elbert, Approaches to neural tissue engineering using scaffolds for drug delivery, *Adv. Drug Delivery Rev.*, 2007, **59**(4–5), 325–338.
- 134 R. Tran, J. Dey and D. Gyawali, Biodegradable elastomeric polymers and MEMS in tissue engineering, in *Biomaterials for MEMS*, ed. M. Chiao and J.-C. Chiao, Pan Stanford Publishing Pte. Ltd, 2011.
- 135 B. Guo and P. Ma, Synthetic biodegradable functional polymers for tissue engineering: a brief review, *Sci. China: Chem.*, 2014, **57**(4), 490–500.
- 136 H. Makadia and S. Siegel, Poly Lactic-co-Glycolic Acid (PLGA) as biodegradable controlled drug delivery carrier, *Polymers*, 2012, **3**(3), 1–19.
- 137 B. C. Gross, J. L. Erkal, S. Y. Lockwood, C. Chen and D. M. Spence, Evaluation of 3D printing and its potential impact on biotechnology and the chemical sciences, *Anal. Chem.*, 2014, **86**(7), 3240–3253.
- 138 S. Lakhundi, R. Siddiqui and N. A. Khan, Cellulose degradation: a therapeutic strategy in the improved treatment of Acanthamoeba infections, *Parasites Vectors*, 2015, **8**(23), 1–16.

- 139 P. L. Nasatto, F. Pignon, J. L. M. Silveira, M. E. R. Duarte, M. D. Nosedá and M. Rinaudo, Methylcellulose, a cellulose derivative with original physical properties and extended applications, *Polymers*, 2015, **7**, 777–803.
- 140 R. Y. Tam, T. Fuehrmann, N. Mitrousis and M. S. Shoichet, Regenerative therapies for central nervous system diseases: a biomaterials approach, *Neuropsychopharmacology*, 2014, **39**(1), 169–188.
- 141 J. M. Aamodt and D. W. Grainger, Extracellular matrix-based biomaterial scaffolds and the host response, *Biomaterials*, 2016, **86**, 68–82, Elsevier.
- 142 J. Dervaux and A. M. Ben, Journal of the mechanics and physics of solids buckling condensation in constrained growth, *J. Mech. Phys. Solids*, 2011, **59**(3), 538–560.
- 143 Q. Zhao, J. Sun, Y. Lin and Q. Zhou, Study of the properties of hydrolyzed polyacrylamide hydrogels with various pore structures and rapid pH-sensitivities, *React. Funct. Polym.*, 2010, **70**(9), 602–609.
- 144 S. Dumitriu, Polymeric biomaterials, revised and expanded, in *High Performance Biomaterials: A Complete Guide to Medical and Pharmaceutical Applications. Illustrate*, ed. M. Szycher, CRC Press, 1991, pp. 824.
- 145 F. Ganji, S. Vasheghani-Farahani and E. Vasheghani-Farahani, Theoretical description of hydrogel swelling: A review, *Iran. Polym. J.*, 2010, **19**(5), 375–398.
- 146 P. M. Budd, B. Mckeown, D. Fritsch, P. M. Budd and P. M. Budd, Free volume and intrinsic microporosity in polymers, *J. Mater. Chem.*, 2005, **15**, 1977–1986.
- 147 S. Yang, K.-F. Leong, Z. Du and C. Chua, The design of scaffolds for use in tissue engineering. Part I. Traditional factors, *Tissue Eng.*, 2001, **7**(6), 679–689.
- 148 R. Deanin, Structure property relations in polyurethanes, in *High Performance Biomaterials: A Complete Guide to Medical and Pharmaceutical Applications. Illustrated*, ed. M. Szycher, CRC Press, 1991, pp. 51–69.
- 149 W.-F. Su, *Principles of Polymer Design and Synthesis Volume 82 of Lecture Notes in Chemistry. Illustrated*, Springer Science & Business Media, 2013.
- 150 P. Davis, S. J. Huang, L. Nicholais and L. Ambrosio, Modified PHEMA hydrogels, in *High Performance Biomaterials: A Complete Guide to Medical and Pharmaceutical Applications. Illustrate*, ed. M. Szycher, CRC Press, 1991, pp. 343–367.
- 151 N. Kamaly, B. Yameen, J. Wu and O. C. Farokhzad, Degradable controlled-release polymers and polymeric nanoparticles: mechanisms of controlling drug release, *Chem. Rev.*, 2016, **116**(4), 2602–2663.
- 152 L. Moroni and J. H. Elisseeff, Biomaterials engineered for integration, *Mater. Today*, 2008, **11**(5), 44–51.
- 153 A. Guiseppi-Elie, Electroconductive hydrogels: Synthesis, characterization and biomedical applications, *Biomaterials*, 2010, **31**(10), 2701–2716.
- 154 M. Morganti-Kossmann, M. Rancan, P. Stahel and T. Kossmann, Inflammatory response in acute traumatic brain injury: a double-edged sword, *Curr. Opin. Crit. Care*, 2002, **8**(2), 101–105.
- 155 R. E. Guldberg, C. L. Duvall, A. Peister, M. E. Oest, A. S. P. Lin, A. W. Palmer, *et al.*, 3D Imaging of tissue integration with porous biomaterials, *Biomaterials*, 2009, **29**(28), 3757–3761.
- 156 Y. Liang, A. Bar-Shir, X. Song, A. A. Gilad, P. Walczak and J. W. M. Bulte, Label-free imaging of gelatin-containing hydrogel scaffolds, *Biomaterials*, 2015, **42**, 144–150.
- 157 T. S. Ali, T. A. Bjarnason, D. L. Senger, J. F. Dunn, J. T. Joseph and J. R. Mitchell, Quantitative T2: interactive quantitative T2 MRI witnessed in mouse glioblastoma, *J. Med. imaging*, 2015, **2**(3), 36002.
- 158 S. Liachenko and J. Ramu, Quantification and reproducibility assessment of the regional brain T2 relaxation in naïve rats at 7 T, *J. Magn. Reson. Imaging*, 2017, **45**(3), 700–709.
- 159 N. J. Luciano, P. Sati, G. Nair, J. R. Guy, S.-K. Ha, M. Absinta, *et al.*, Utilizing 3D printing technology to merge MRI with histology: A protocol for brain sectioning video link, *J. Visualized Exp.*, 2016, **118**, e54780.
- 160 R. Vink, Large animal models of traumatic brain injury, *J. Neurosci. Res.*, 2018, **96**, 527–535.
- 161 H. M. Kaplan, P. Mishra and J. Kohn, The overwhelming use of rat models in nerve regeneration research may compromise designs of nerve guidance conduits for humans, *J. Mater. Sci. Mater. Med.*, 2015, **26**(8), 1–5.
- 162 H.-W. Han and S.-H. Hsu, Using 3D bioprinting to produce mini-brain, *Neural Regener. Res.*, 2017, **12**(10), 1595–1596.
- 163 M. W. Wojnarowicz, A. M. Fisher, O. Minaeva and L. E. Goldstein, Considerations for experimental animal models of concussion, traumatic brain injury, and chronic traumatic encephalopathy-these matters matter, *Front. Neurol.*, 2017, **8**, 240.
- 164 V. E. Johnson, D. F. Meaney, D. K. Cullen and D. H. Smith, Animal models of traumatic brain injury, *Handb. Clin. Neurol.*, 2015, **127**, 115–128.
- 165 K. E. Saatman, A.-C. Duhaime, R. Bullock, A. I. R. Maas, A. Valadka and G. T. Manley, Classification of traumatic brain injury for targeted therapies, *J. Neurotrauma*, 2008, **25**(7), 719–738.
- 166 S. Lee, M. Nowicki, B. Harris and L. G. Zhang, Fabrication of a highly aligned neural scaffold via a table top stereolithography 3D printing and electrospinning, *Tissue Eng., Part A*, 2016, **23**(11–12), 491–502.
- 167 P. Z. Elias and M. Spector, Characterization of a bilateral penetrating brain injury in rats and evaluation of a collagen biomaterial for potential treatment, *J. Neurotrauma*, 2012, **29**(11), 2086–2102.